





Fundamentals of Machining

↳ WHAT IS
MACHINING!

Groover M.P.: Chapter 17



LEARNING TARGETS

2

principles of ^{common} machining operations and tools

↓
machine tools + needed device

The student knows how:

- understand the mechanism of chip formation;
- knows industrial machining processes;
- optimize costs in machining;
- improve process planning. ← how to plan machining operations



Machine tools
with numerical control
→ (elements of CNC machine)
different NC kind --- tool

↓
numerical control in manufacturing

↓
(similar to coord meas. syst machine casting, metal forming...)





MACHINING PROCESSES

perform to add quality → dimension/Geometry / surface finish → variability inside a given range
(VARIABILITY OF FINAL PART)
How to plan operation } (also for surface finish)
to RESPECT requirements } What is ROUGHNESS

3

removed material from WORKPIECE to obtain the final OBJECT → FOR A BETTER Quality we perform machine ...

Goal: to remove material to transform the raw part into the desired geometry.

In general they are the last processes performed on the mechanical components. In fact, machining processes allow to obtain good tolerances and good surface finish.



A forged crankshaft before and after machining the bearing surfaces. The shiny bearing surfaces of the part on the right cannot be made to their final dimensions and surface finish by casting or metal forming processes.



MACHINING PROCESSES

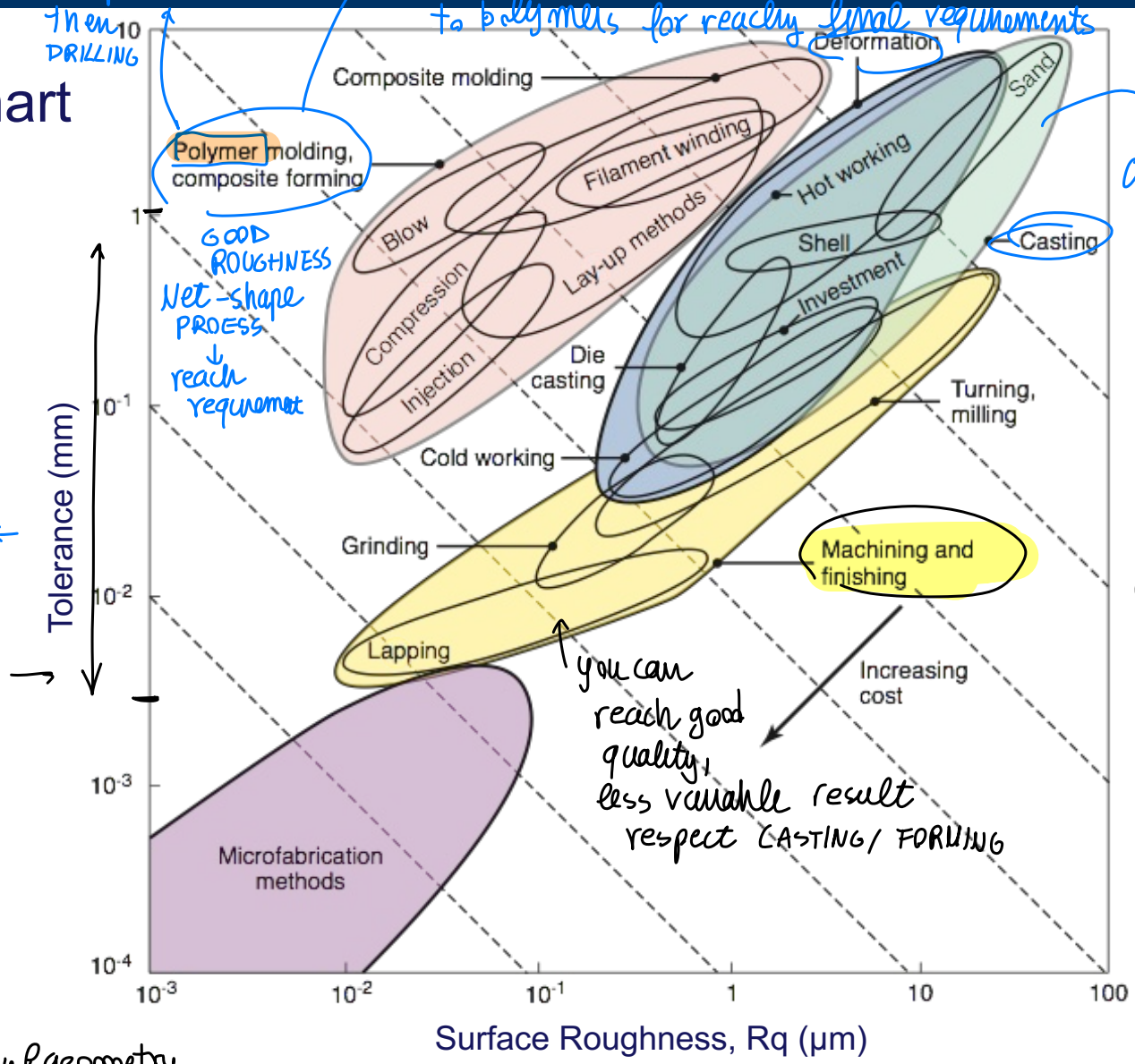
similar to machine, but **Polymer process** 4
are **NET-SHAPE** → but machining can be applied also to polymers for reaching final requirements
to have particular geom features

Ashby Chart

all operations can be done also on polymers

{ like the size }

we can reach a good quality in terms of tolerances / surface
machines reach final requirements and CAST/FORM for initial geometry



Casting / metal forming are overlap

All machining operation can be done on polymers

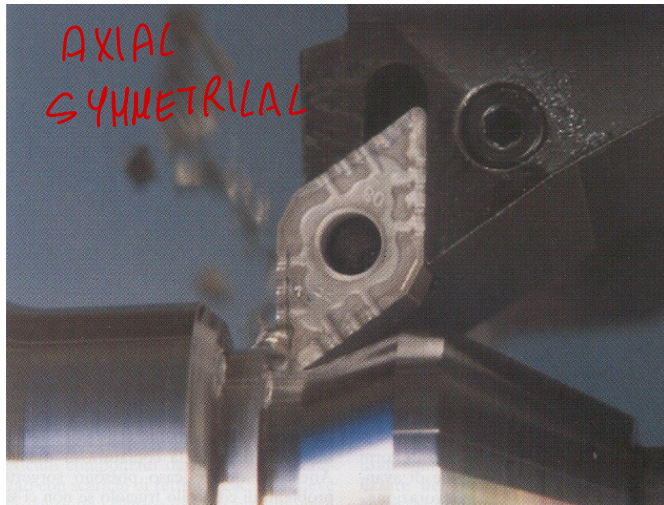


MACHINING PROCESSES

5

basic operations of manufacturing environment

INDUSTRIAL OPERATIONS



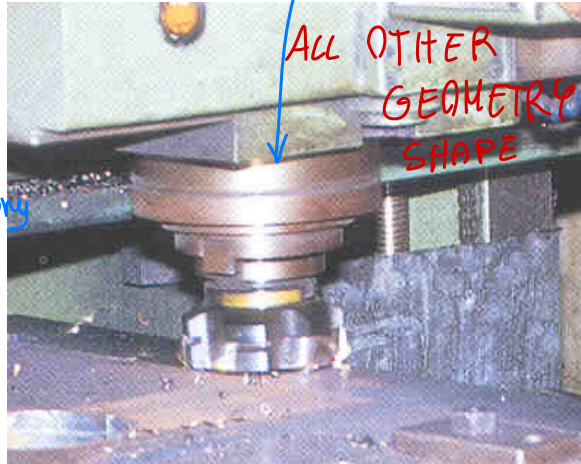
Turning

obtain **AXIAL** symmetrical shape features

by ROTATION of the parts...
line trajectory of the tool to have ax. sym.

different configuration for complex surface finish

from a BAR, obtain axial sym features



Milling

all other possible geometries requirements you can find on parts

Drilling

JUST to make holes (cylindrical or complex)
(Hole making TWIST DRILLING + others)



very complex tool motion...
PRISMATIC / COMPLEX FEATURE

different configuration
↑
ROTATORY TOOL, motion of the tool and workpiece
↑



MACHINING PROCESSES

We will study: *simple INDUSTRIAL PROCESS*

- ✓ **“Orthogonal” cutting**
a “simplified” industrial process

*Situation considered because simpler,
described when all kinematics are all
on one plane* ⇒ chip formation basis
relevant for metal removal

- ✓ **More complex industrial processes:**

- Turning
- Drilling
- Milling
- etc.

*↓
“CHIP” is the
metal removed
from the part*



We study this machining process because:

- It is **simple** to describe from the kinematic (motions) and dynamic (forces) points of view;
- It allows us to **understand** the elementary mechanism of chip formation;
- Many of the **variables** met in orthogonal cutting are present in the more complex industrial processes (turning, milling, etc).

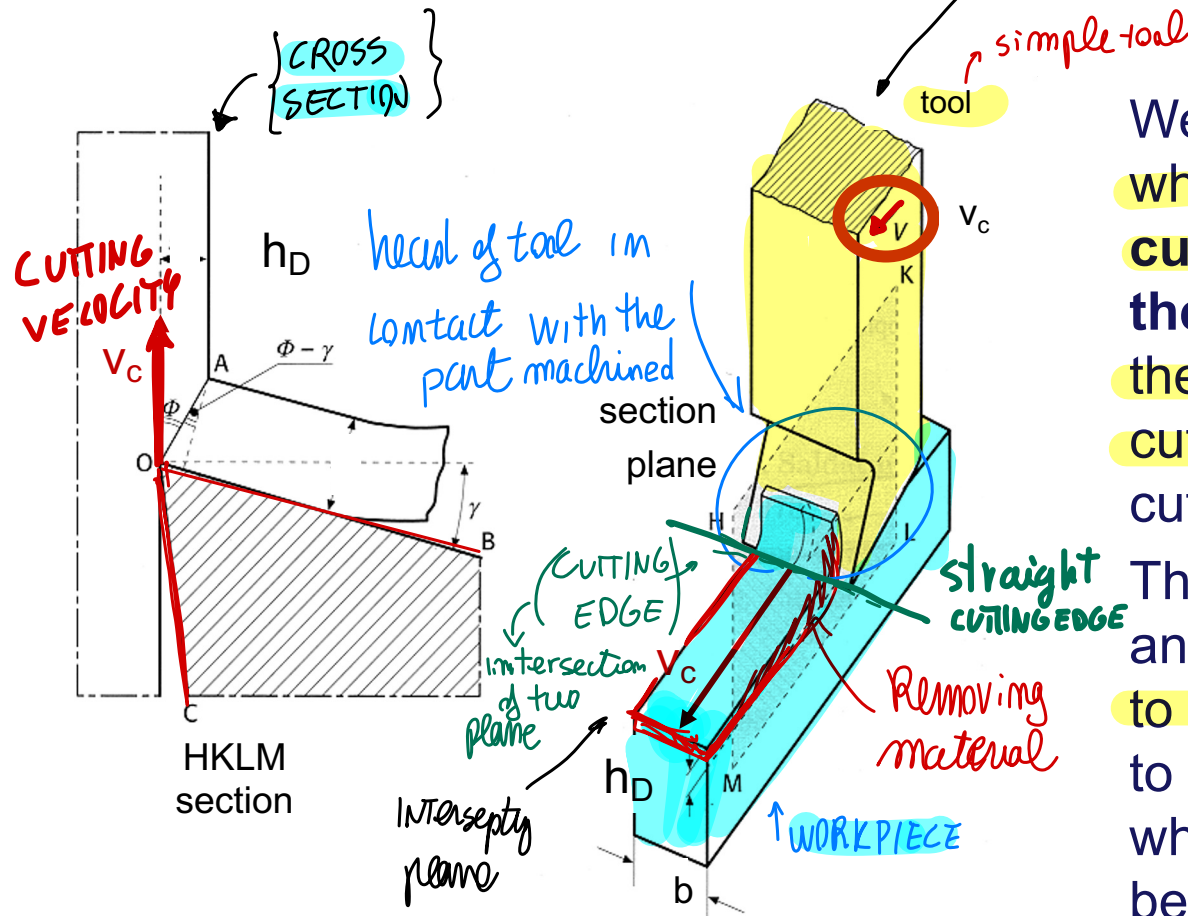
} all on a plane!



ORTHOGONAL CUTTING

definition of this situation process
In which the tool is this

8

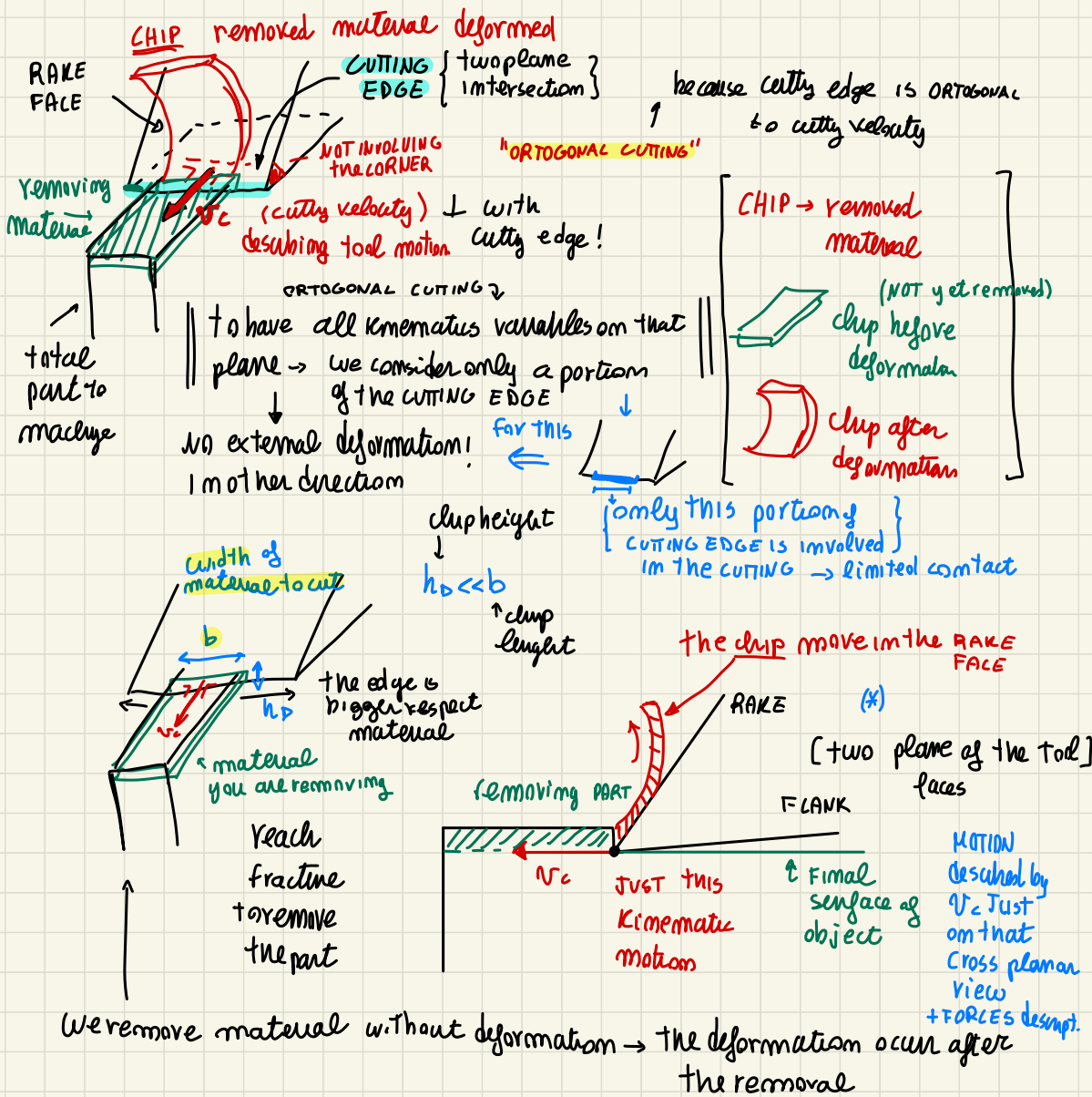


We have orthogonal cutting when the tool moves with a cutting speed orthogonal to the cutting edge and only the central portion of the cutting edge is involved in the cutting.

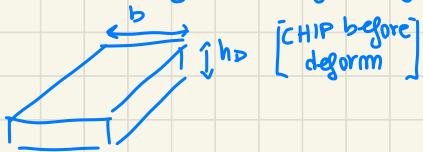
Therefore, all the kinematic and dynamic variables belong to a plane that is orthogonal to the cutting edge and to which the cutting speed belongs.

CUTTING } Intersection of
EDGE } two plane $\rightarrow$ sharp edge

Position the tool so his motion (described by V_c) is ORTHOGONAL to the cutting edge



h_d : dimension of chip before deformation (thickness)



usually $h_d \ll b \dots$

contact with material, with deformation until fracture to remove material

IF this condition respected...

we can describe the process in a PLANE ORTHOGONAL TO CUTTING edge (CROSS section view) (*)

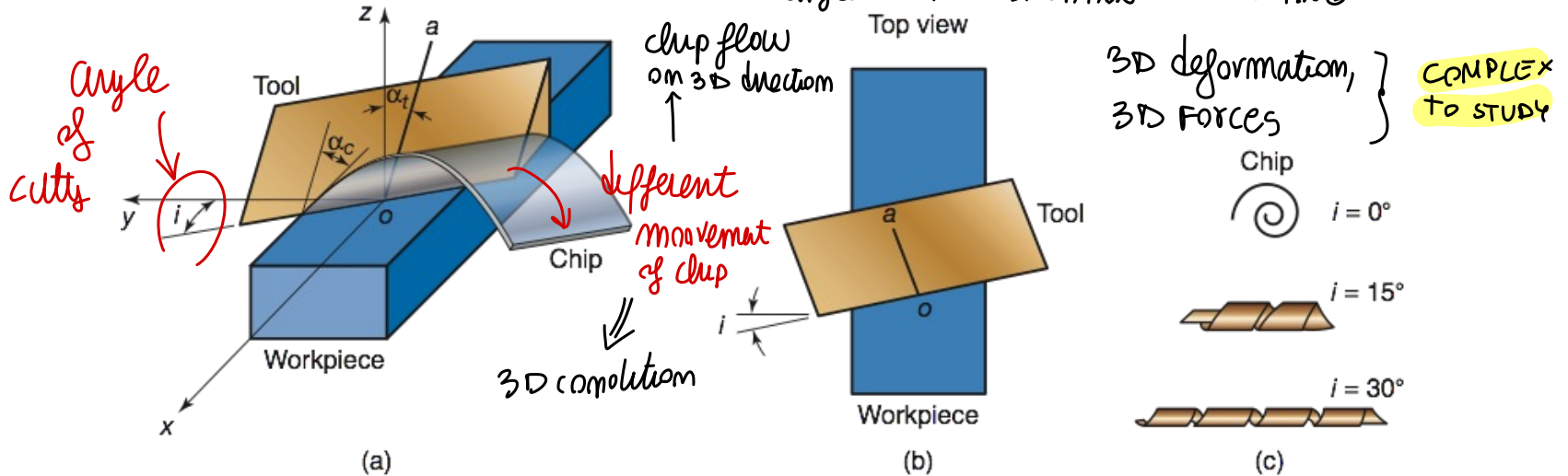


CUTTING WITH AN OBLIQUE TOOL

{ NO MORE ORTHOGONAL V_c to cutting edge! }⁹

JUST changing something... modify direction of CUTTING EDGE respect cutting speed
In practice, most of the cutting processes use oblique tools and the study of chip formation is quite complex.

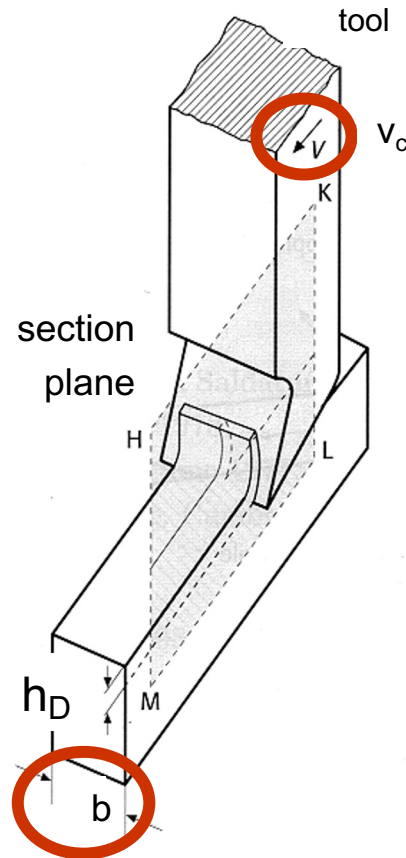
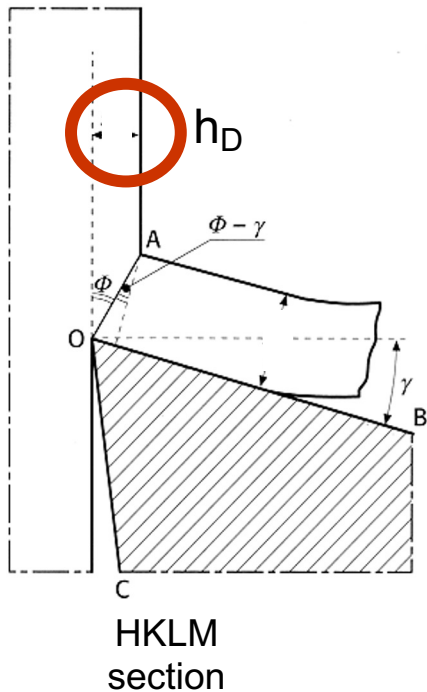
$\hookrightarrow$ in other industrial process (general cutting) $\rightarrow$ generalization of ORTHOGONAL CUTTING
we introduce a small angle $i \rightarrow$ NEW SITUATION



if you modify the cutting edge direction respect cutting speed $\rightarrow$ with small angle i , situation change

$\Rightarrow$ chip move on 3D

Real situation in INDUSTRIAL PROCESS
(generalization of ORTHOGONAL CUTTING)



Main variables to define!

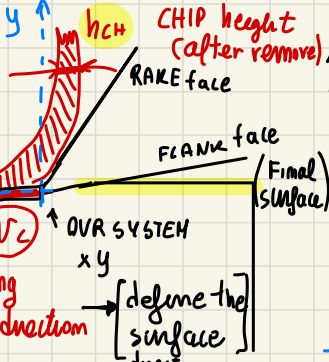
Relevant Variables:

- $v_c = \text{cutting speed}$ $\perp$ cutting edge
- $b = \text{cutting width}$
- $h_D = \text{cutting thickness}$
(chip thickness before chip separation)

LD

in which we describe everything

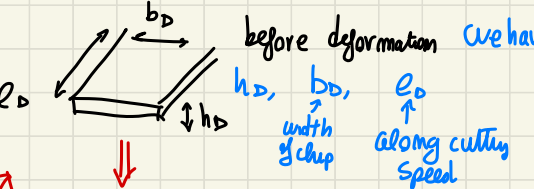
Reference system



h_D : before deformation thickness of chip
 h_{CH} : after deformation of chip

we can define area of material we are removing before deformation
 $A_D = h_D \cdot b_D$ Area of chip before deformation

ORTOGONAL ALWAYS! theoretical area of material of approach



(after deformation) h_{CH}, b_{CH}, l_{CH}

Because it is a Deformation process $\rightarrow$ VOLUME IS CONSTANT
 $V_D = V_{CH}$
 $h_D b_D l_D = h_{CH} b_{CH} l_{CH}$

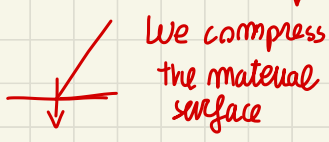
if all kinematics from a plane $\rightarrow$ NO WIDTH change
 $b_D = b_{CH} = b$
 NO REASON to deform b {ORTOGONAL deform}

Same width, doesn't change
 JUST CHANGES on one plane..

we define: chip thickness ratio

$$r_c = \frac{h_D}{h_{CH}} < 1$$

usually we COMPRESS material ($h_{CH} > h_D$) to lead it to fracture so we have

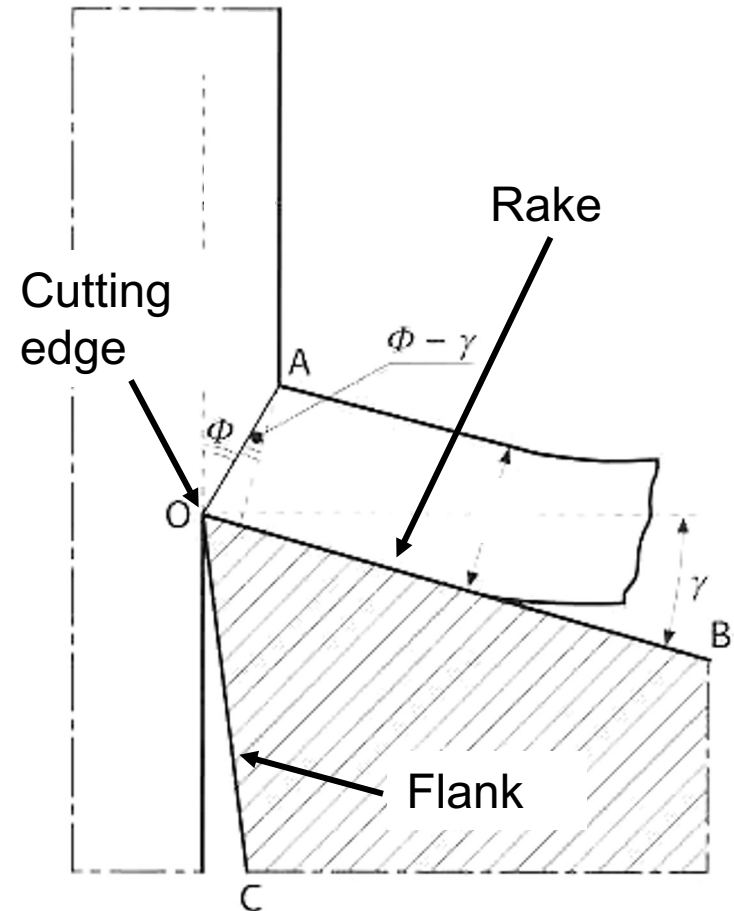


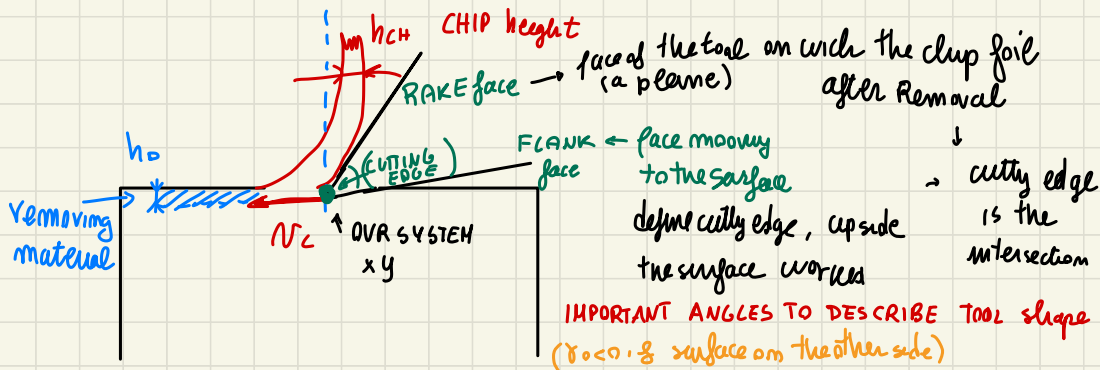
We compress the material surface



The tool is composed of:

- **Rake face:** surface on which the chip flows.
- **Flank face:** surface looking at the machined surface.
- **Cutting edge:** intersection line between rake face and flank.



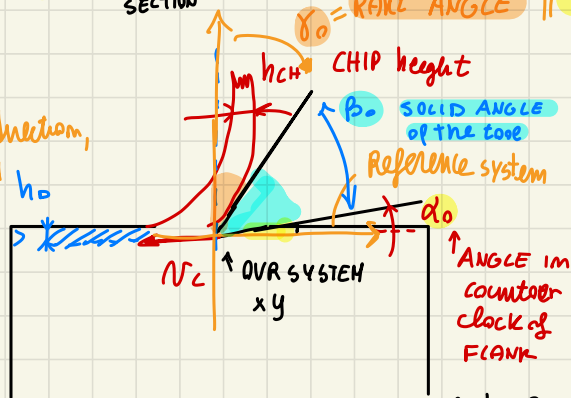


|| **main angles** || along CROSS-SECTION

$\gamma_0 > 0$ measured clockwise respect vertical axis

$\gamma_0 = \text{RAKE ANGLE}$ || $-15^\circ \leq \gamma_0 \leq 30^\circ$ || typical range

both ≥ 0 values vary angle depending on the MATERIAL that you have to cut



$2^\circ \leq \alpha_0 \leq 15^\circ$

MUST BE POSITIVE, otherwise

There is a sliding PROBLEM

creating other compression on the surface IF $\alpha_0 < 0$

constant compression on material $\Rightarrow$ damage tool and surface!

lot of loss $\Rightarrow$ Temperature increases energetic loss!

constant sliding and compression on material yet cutted $\Rightarrow$ damaging the tool and surface

$$\alpha_0 + \beta_0 + \gamma_0 = \pi/2$$

If $\alpha_0 \uparrow$, and γ_0 fix $\rightarrow \beta_0 \downarrow$

Solid angle reduction $\Rightarrow$ SHARPER BUT less ROBUST!

usually α_0 @ MIN and we play with γ_0, β_0

$\Rightarrow$ if material easy to cut compression until fracture! $\rightarrow$ on material

α_0 chosen according to material to cut $\rightarrow$ easy to cut material moving δ_0 ..

α_0 is also important IF fixing δ_0 .. $\alpha_0 > 0$ for sure

having BIG $\alpha_0 \rightarrow \beta_0$ reduces

less $\beta_0 \Rightarrow$ sharper tool
BUT less robust



usually we set β_0 to minimum (enough robust)
and play on value of δ_0 according on different situation

• easy to cut material: cutting process $\equiv$ deformation process
(compression, deform until fracture)

(easy to deform
and bring to
fracture)

$\rightarrow$ less force needed

$\hookrightarrow$ so the tool can be less robust
(less force) -- we try to have
the biggest $\alpha_0 \uparrow$ and $\delta_0 \uparrow$

α_0 fixed $\delta_0 \uparrow \rightarrow$ sharper tool



less robust BUT easy cutting

easy to cut $\leftrightarrow$ easy to deform

$\downarrow$ less FORCE needed $\Rightarrow$

even if β_0 sharper
less ROBUST

$\downarrow$

sharper tool $\Rightarrow$ easy to cut material

enough Robust but very sharp

So $F \downarrow$ to fracture (cut)

|||

tool can be less ROBUST

If easy to cut...

you try to have β_0 high $\uparrow$

to FIX, $\beta_0 \downarrow \Rightarrow$ SHARPER

less forces to cut if

$\checkmark$ you have sharper tool

$\downarrow$ DIFFICULT TO CUT Material

• || of HARD MATERIAL || $\rightarrow$ ROBUST TOOL needed

to not damage the tool

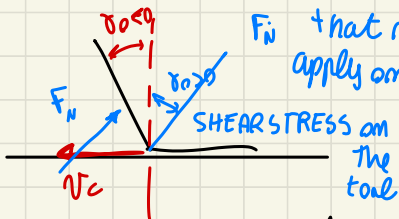
so material of tool harder than

CUTTING one $\rightarrow$ but harder means

more brittle material

to avoid this we consider an $\beta_0 < 0$

$\checkmark$ { also the tool } $\downarrow$
CAN FRACTURE!



F_n that material apply on tool

hard material resist more to compression

HARD tool material

- $\beta_0 > 0 \rightarrow$ shear less resistance
- $\beta_0 < 0 \rightarrow$ compression more resistance of tool

(we need)

MODELS to

prove that using an HARDER tools, compression stress is more resistant than shear

$\downarrow$
+ models that describe forces, chip formation, so on..

{ The component of forces that material apply on tool is compression
HARD material resist better to compression than shear stress!

if we consider the force F_n orthogonal to the surface..



Cutting angles: *Important angles that you can define →* cross section view

- **Rake angle γ_0**

Between the rake face and the normal to the cutting direction:

$$-15^\circ \leq \gamma_0 \leq 30^\circ$$

- **Clearance (or relief) angle α_0**

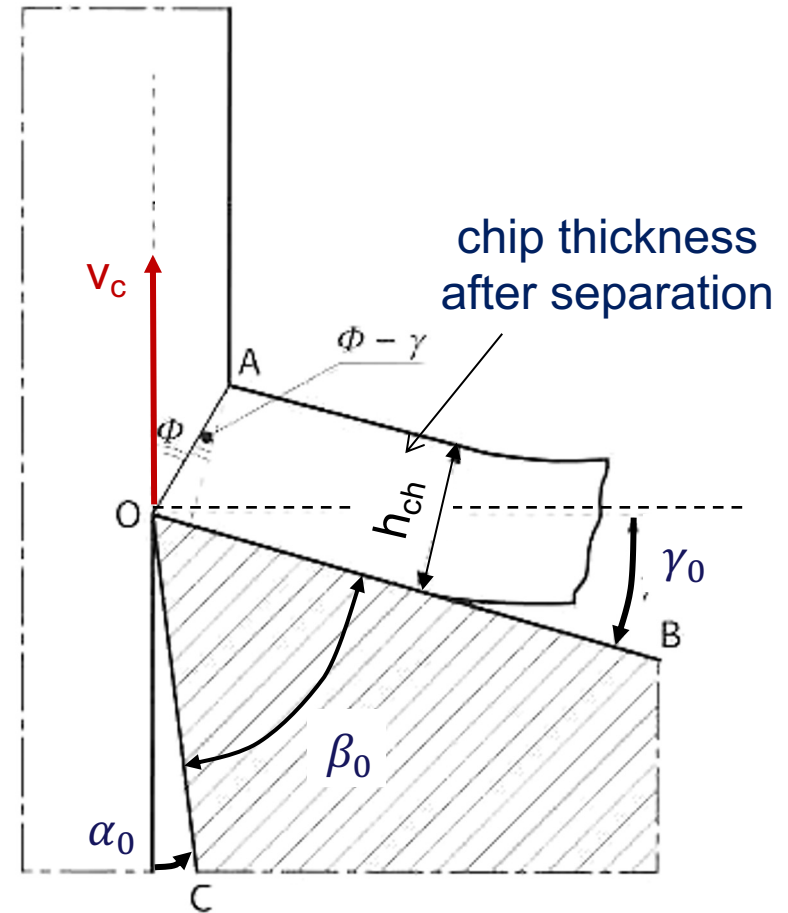
Between the flank and the cutting direction:

$$2^\circ \leq \alpha_0 \leq 15^\circ$$

- **Solid angle β_0**

Between rake and flank faces:

$$\alpha_0 + \beta_0 + \gamma_0 = 90^\circ$$





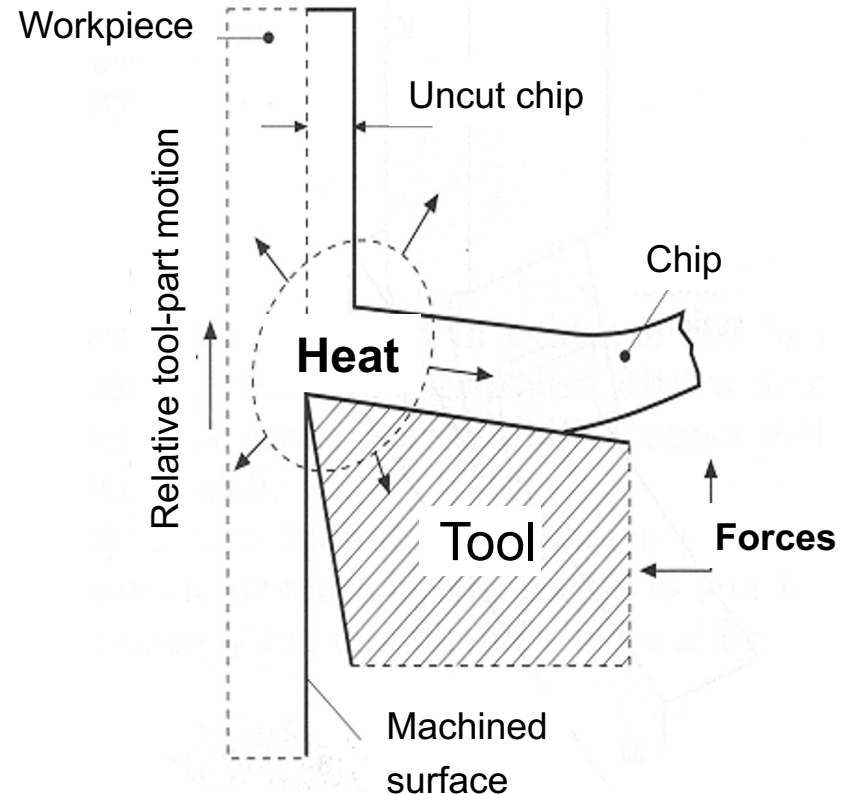
MECHANISM OF CHIP FORMATION

- A **relative motion** between tool and workpiece exists.
- The **tool stresses the material**. *ELASTIC DEF.*
- The material is **plastically deformed to fracture**, separates from the workpiece and is transformed into chip.
- The **chip flows** on the tool and moves away from the machining zone.

IDEAL
SITUATION...

↳ we
have Forces
+ Heat

Cutting process





MECHANISM OF CHIP FORMATION

Associated to chip removal:

- **Forces:** Tool and workpiece exchange forces needed to deform the material, separate it from the workpiece and transform it into chip.

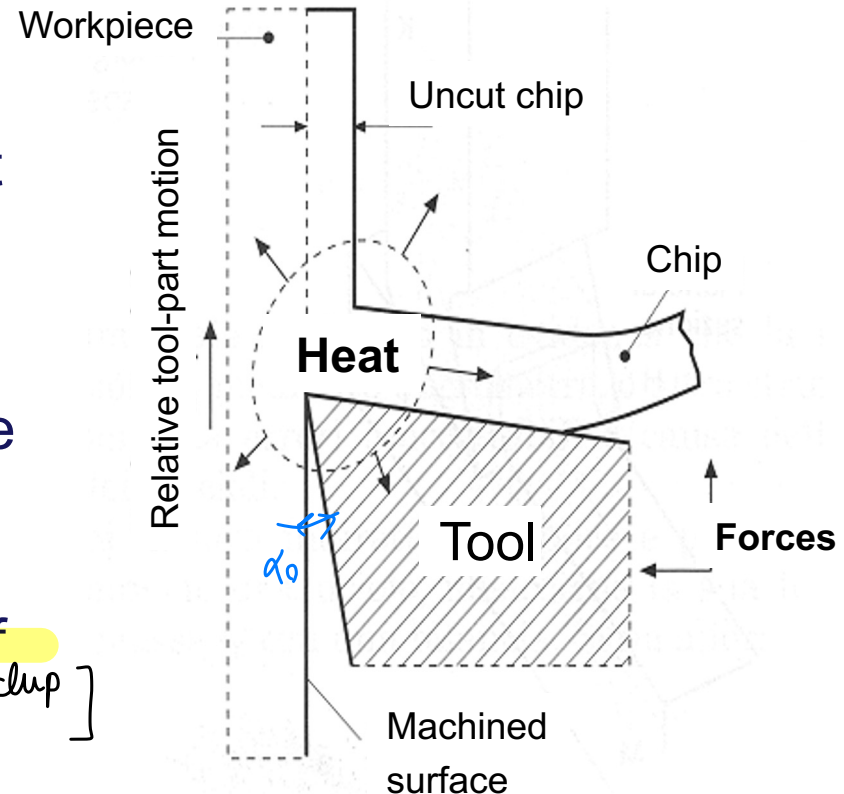
- **Heat:** Plastic deformation of the material before chip separation and friction between chip and tool generate a large amount of heat.

[also Friction of chip
on RAKE FACE]

↓ due to deformation process and FRICTION
we have heating energy

$\alpha_0 > 0$ to reduce
FRICTION ecc..

Cutting process



While the material is deformed, we have
ELASTIC RECOVERY $\rightarrow \alpha_0 > 0$ so there is
space for elastic recovery

ideal description until now...

obviously, when breaking material, when the tool
surpass the material surface $\rightarrow$ we have ELASTIC RECOVERY

In fact we have
to choose $\alpha_0 > 0$



$\swarrow$ little changes on final
surface..



so that after machining
surface, there is no sliding } less heat
sliding..



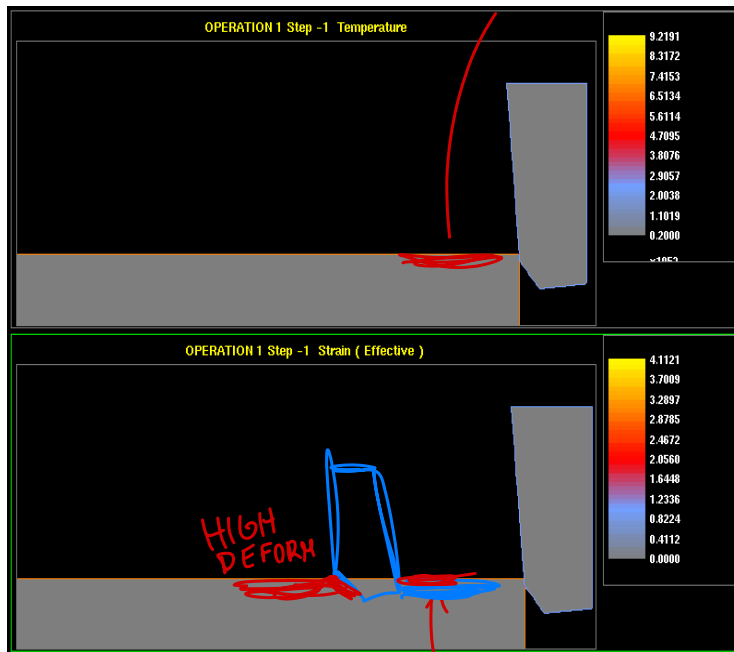
STRESS AND TEMPERATURE

16

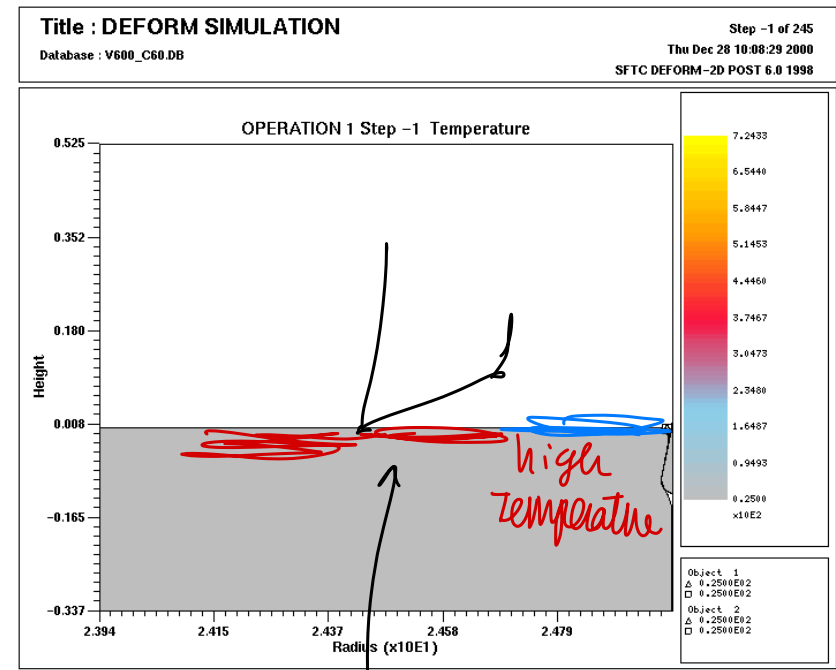
longer heats of surface after the tool surpass, still heat

temperature

strain



less deformation after recovery

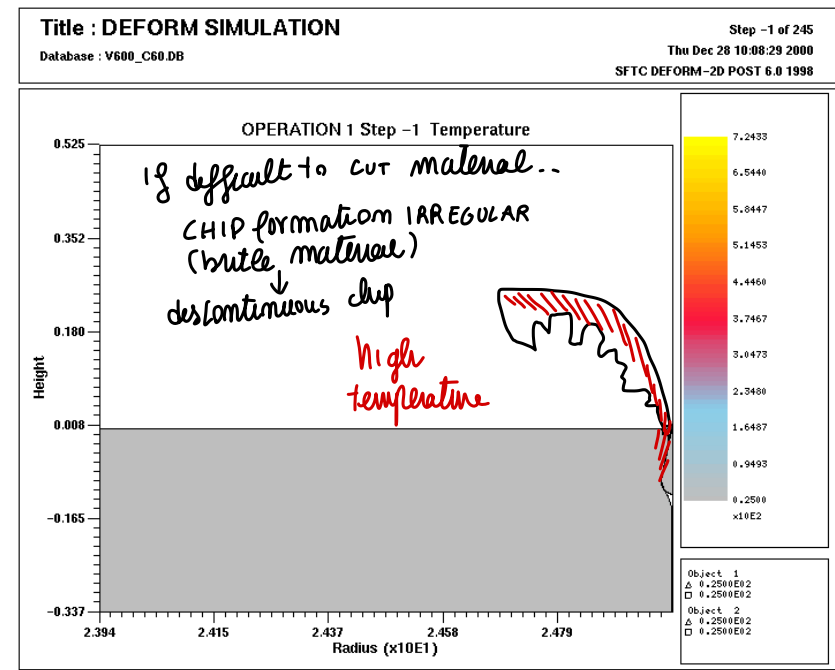
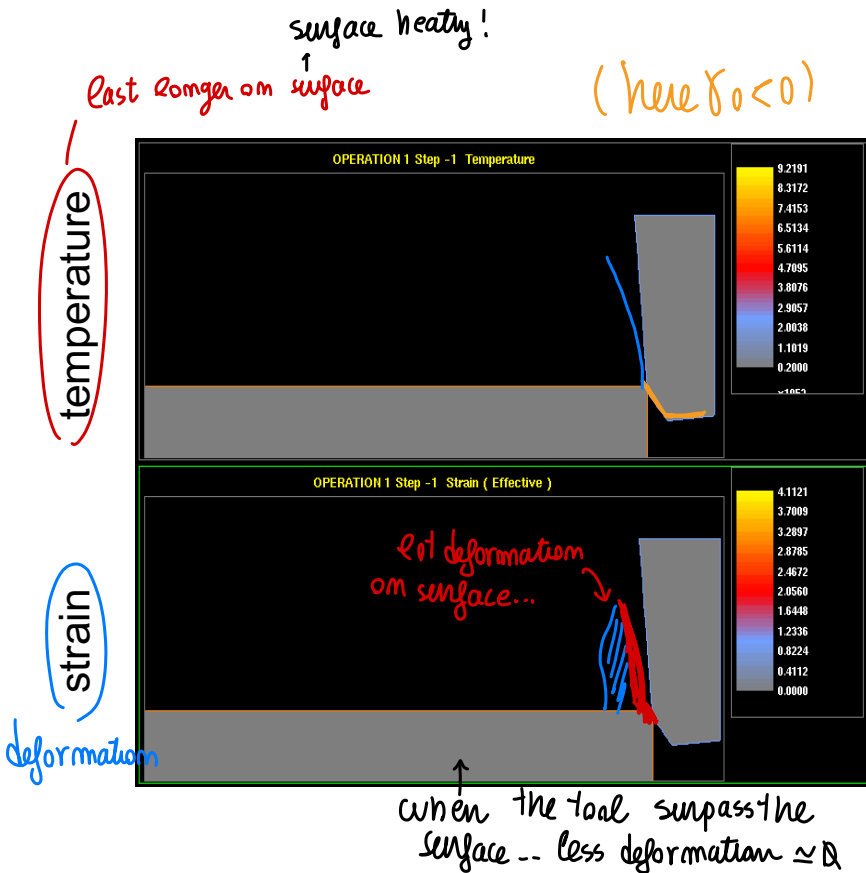


↑ typical of BRITTLE MATERIAL!
In a difficult cutting material, IRREGULAR CUT



STRESS AND TEMPERATURE

16

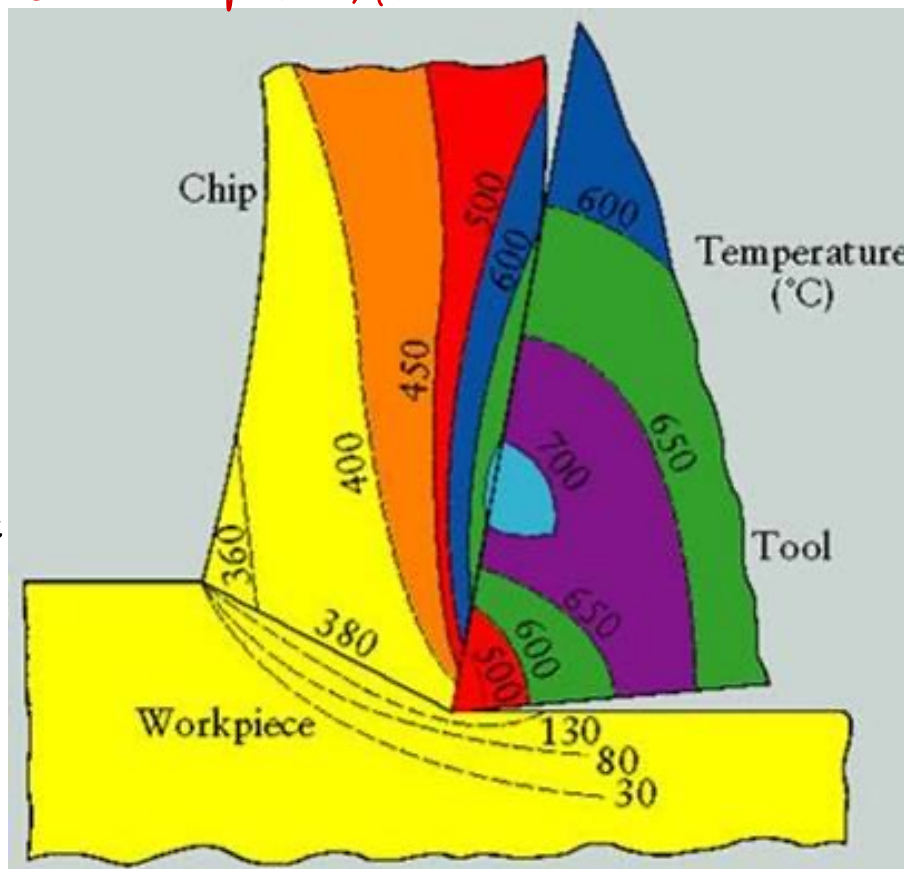




TEMPERATURE [ORTOGONAL CUTTING]

(ISSUE)

due to compression, fracture



You try
to reduce
needed
lubricant to,
less impact
on environment

↓
relevant
Temperature

High Tool
Temp.

Affecting
tool's wear

Dang chip
formation,
UP to

700 °C

↑
HIGH TEMP!
during machine
usually we
use

- coolant to keep not too high Temp
- lubricant to remove friction

↑
material liquid
to add to the part,
{OIL & WATER} MIX
to manage... → try to have

← additional material of the process --
FLUID that impact the system → MUST be well used ---
(or reduce how much of this we need)



MECHANISM OF CHIP FORMATION

DEFORMATION PROCESS

↓
we look the shear stress effect

How does the material (machining allowance) transform into chip?

The tool stresses the material until this deforms plastically. With good approximation it can be said that the region where the material is deformed is a plane, called **shear plane**.

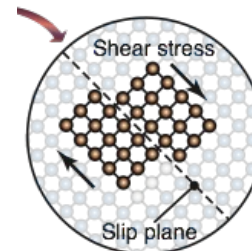
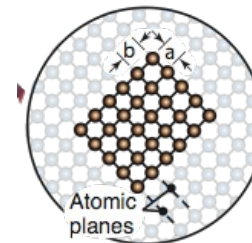
The deformation proceeds until the separation of the chip from workpiece. Then, the chip flows on the tool.

on Metal CUTTING,
How shear happens..

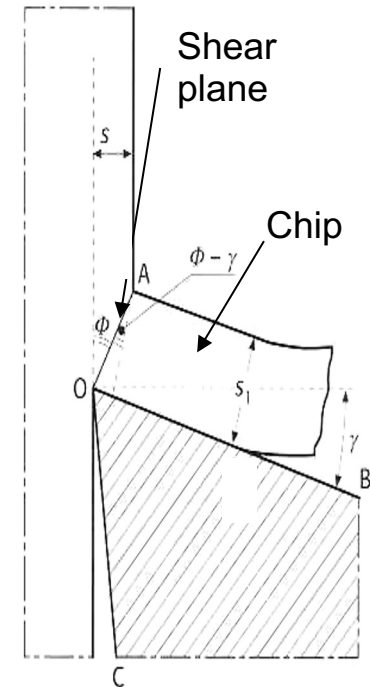
Metal CUTTING → how it happens
to shear strain

describe deformation
due to SHEAR STRESS → nearly def. up to fracture

CRISTALINE
STRUCTURE of material (Metals)



Shear stress
↓
deformation





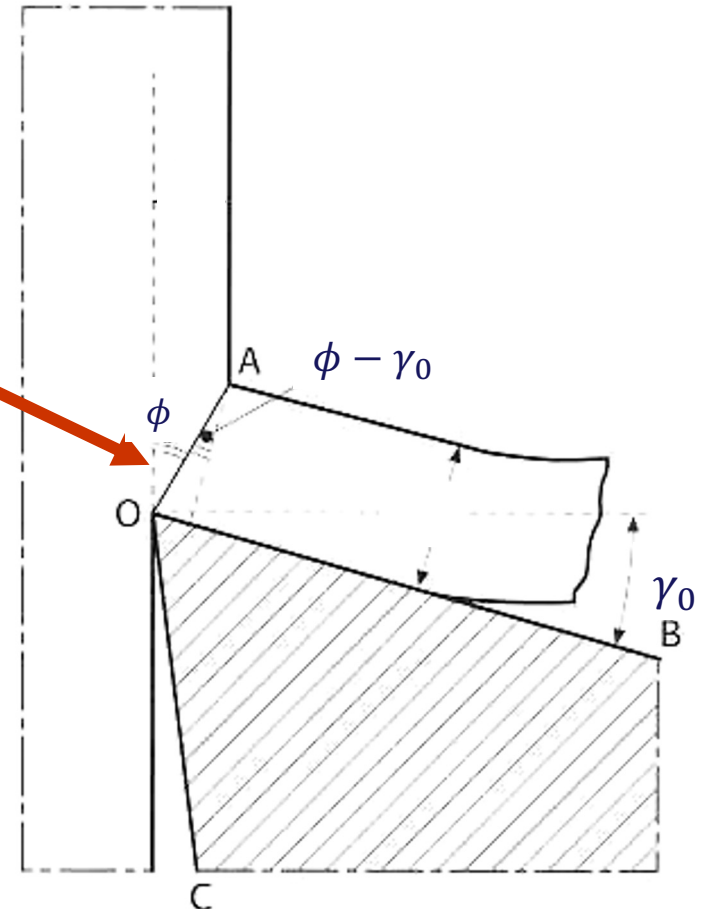
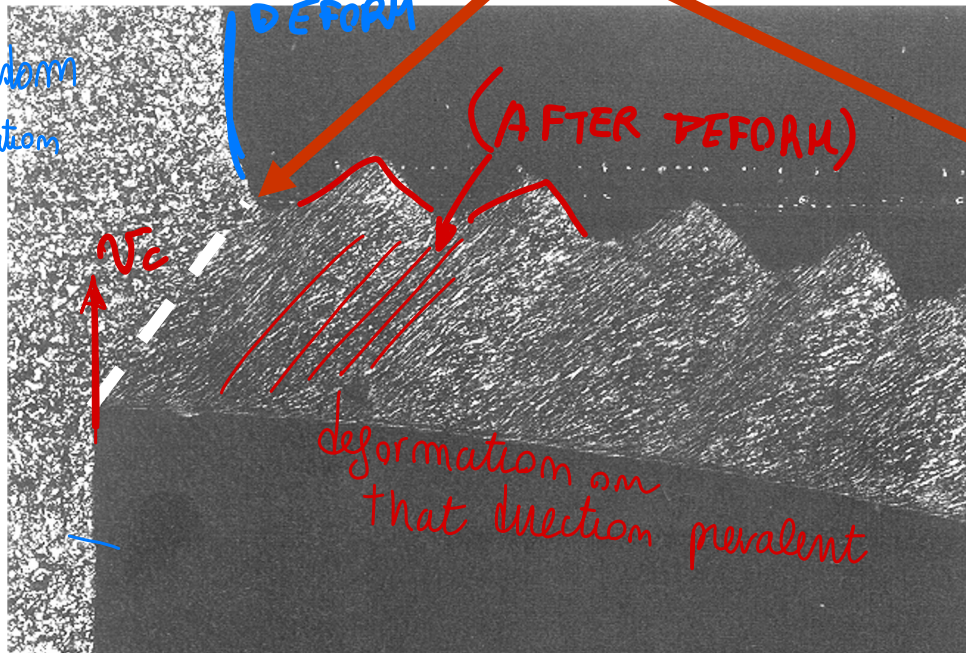
MECHANISM OF CHIP FORMATION

{ ORTHOGONAL }
CUTTING

BEFORE
DEFORM

Shear plane

Random
distribution



deform of crystalline grains of the structure ...
⇒ usually we characterize chip formation considering a zone where shear happen

SHEAR ZONE
≈ simplified as a
surface = PLANE

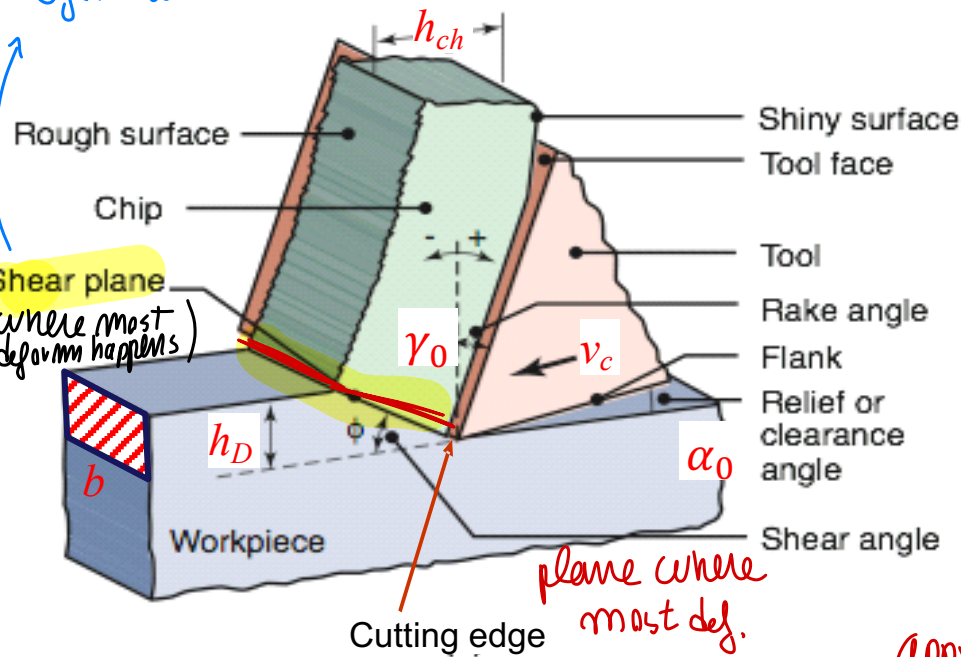


MECHANISM OF CHIP FORMATION

pm a 3D VIEW

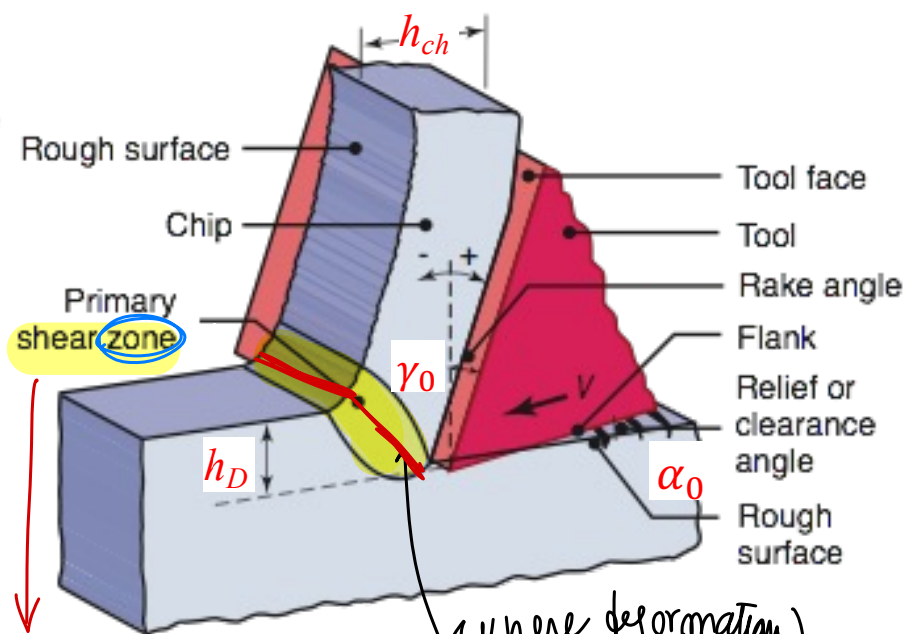
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3D Region where deformation shear happens



Shear plane model

direction respect cutting tool motion in which most deformation happens --



Shear zone model

approx as a surface plane

(where deformation happens)



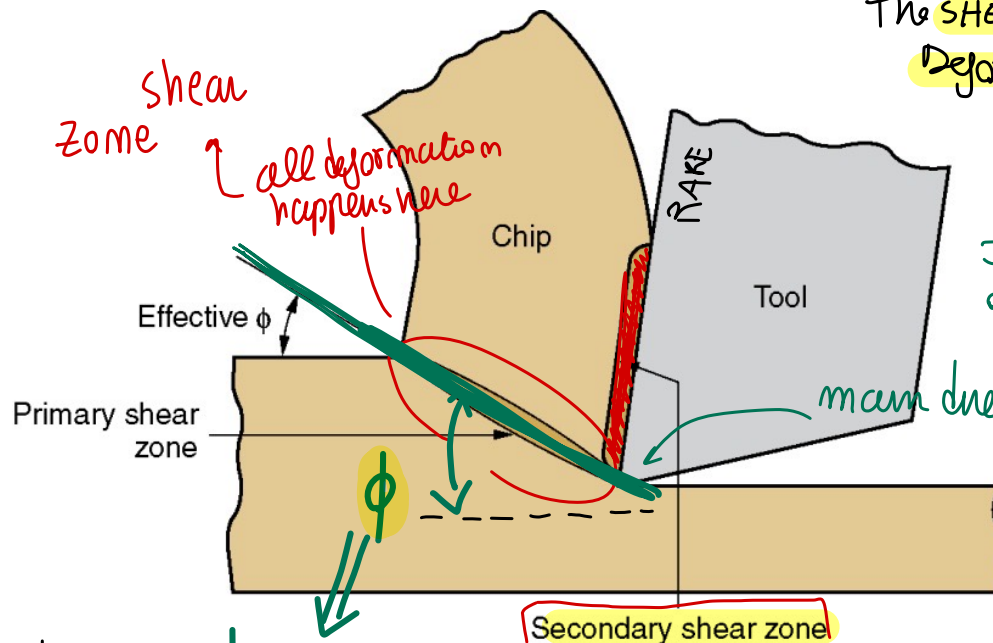
MECHANISM OF CHIP FORMATION

More realistic view of chip formation, showing shear zone rather than shear plane. Also shown is the secondary shear zone resulting from tool-chip friction.

We SIMPLIFY the representation, considering the SHEAR PLANE as cell deformation!

CHIP FORMATION

Just consider all deformation on a simple PLANE!
↑
main direction of that zone -- (shear plane)



angle
to understand
shear
DIRECTION
(where deform
happens)

(shear
angle)
direction respect
cutting speed

Secondary shear zone

↓ due to the friction of
material after removal
flowing on RAKE face

↑ after removal
effect! ↓
relevant! sliding and
stresses influence the
TOOL and tool life



MECHANISM OF CHIP FORMATION

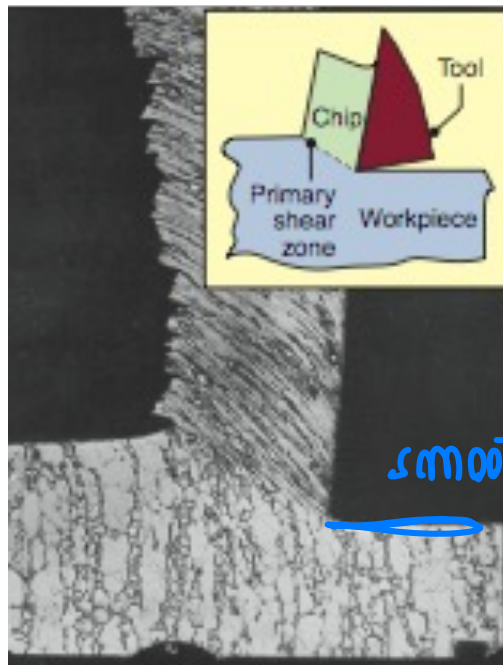
Continuous chip

for an easy to cut material

separate
material
as continuous
chip

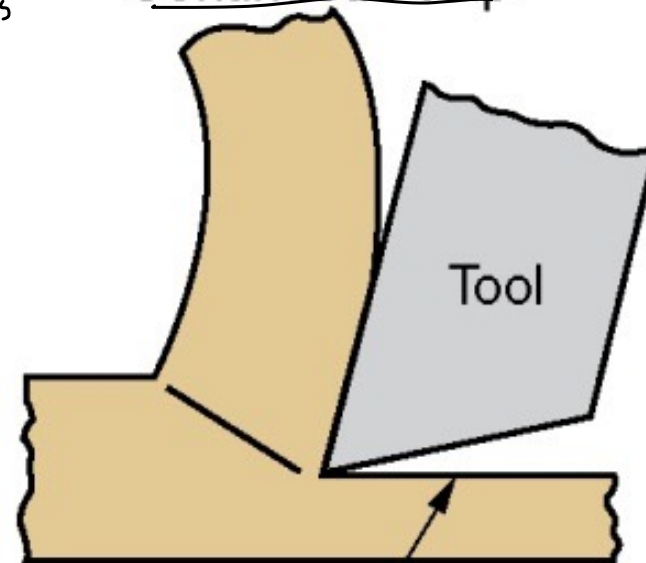
↓ material
removed

Continuous chip



(easy to cut)

smooth final
SURFACE



Good finish typical

↑
obstacle chip,
difficult to Remove!

You have to stop
PROCESS and
remove it..



MECHANISM OF CHIP FORMATION

Discontinuous chip (brittle materials)

hard to cut

divide chip in small pieces

Discontinuous chip

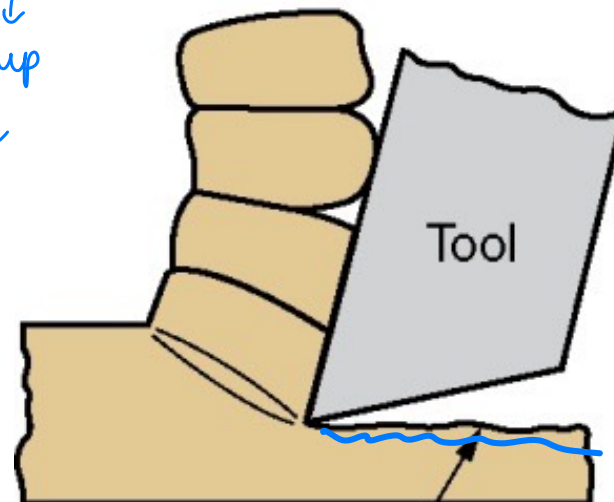


chip broken in small pieces

break chip in small pieces

more irregular final surface

better for chip removal

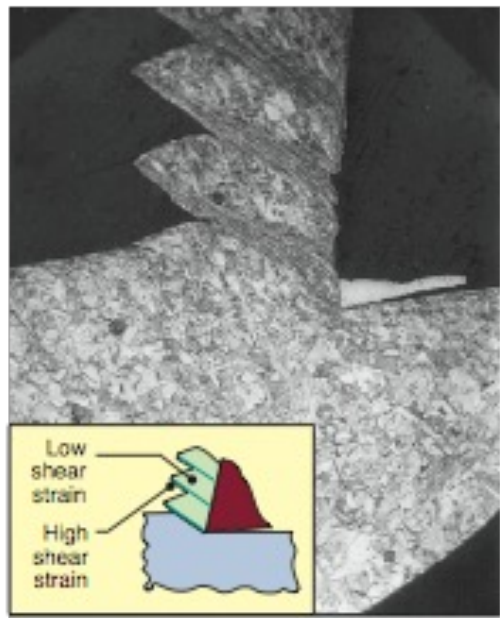


Irregular surface due to chip discontinuities
final surface IRREGULAR (hard to cut!)



MECHANISM OF CHIP FORMATION

Serrated or segmented chip: IN BETWEEN hard/easy to cut
semi-continuous chip with saw-tooth appearance



material is subjected
to different STRAIN

depends
on the
position

High shear
strain zone

Low shear
strain zone

[PARTIALLY
CONTINUOUS/
DISCONTINUOUS]

↓ hybrid condition



- IF IN A PROCESS you are able a continuous chip or discontinuous broken
 - DISCONTINUOUS better for chip removing, you have to remove the chip, if it remain, it is an obstacle! removing chip it is easy to remove if DISCONTINUOUS
- Which Better?

CONTINUOUS

- lot chip that create obstacle, CHIP must be removed by hand

vs

DISCONTINUOUS (CHIP)



- better in removing the chip after deformation -- remove respect cutting Area

IF chip remains on cutting area it is

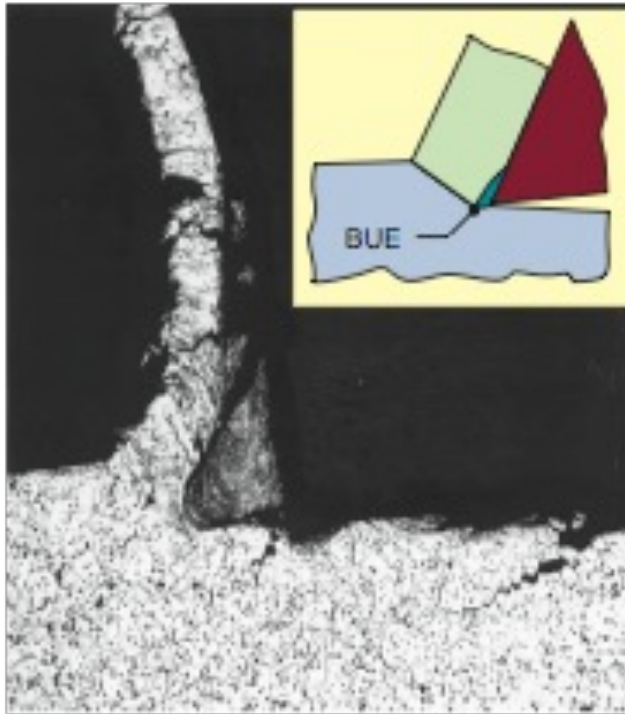
an obstacle → if it breaks in pieces, easy to remove from working area

↓ CNC machy ⇒ remove chip --

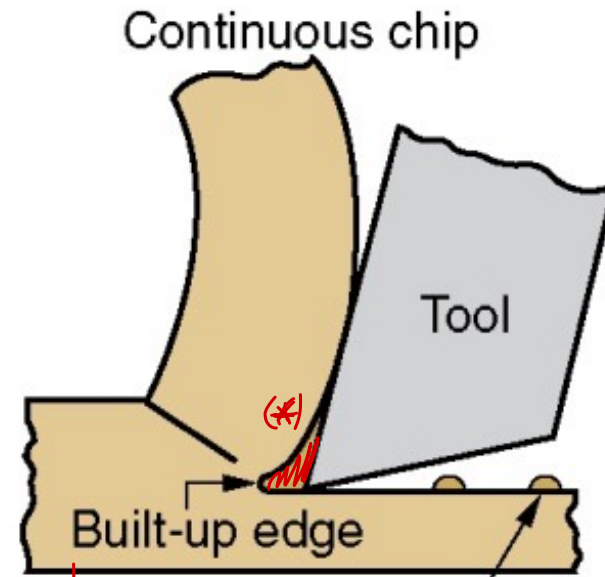
chip lubricant -- separate chip that can be re-used with lubricant --



Continuous chip with built-up edge (BUE) (*)



adhesion of material due to Forces and heat → adhesion on tool → built-up material on the cutting edge → **UNDESIRABLE!**



due to FORCES, HEAT, adhesion of material, built up material on CUTTING Edge → **DON'T LIKE IT!** ⇒



MECHANISM OF CHIP FORMATION

Continuous chip with built-up edge (BUE)

you need to have higher V_c to avoid that!



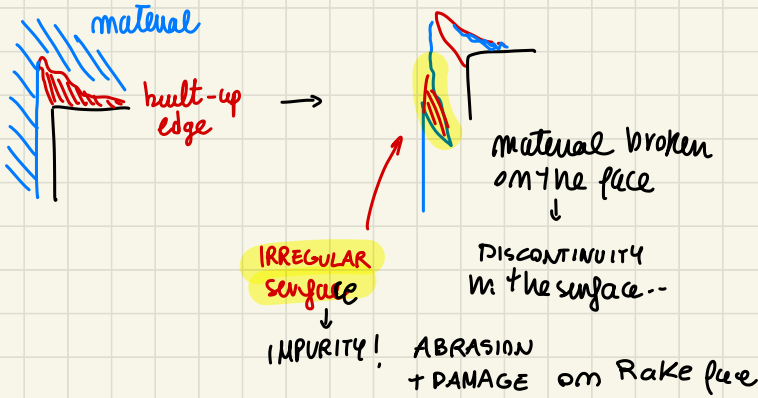
IT can create SURFACE
DISCONTINUITY

Damaging
surface and
tool → MUST NOT HAPPEN

Not well controlled
CUTTING! ↓ it depends on
cutting speed V_c

BUE you start cutting with a deformed FLANK face--

↓
changing cutting edge shape--



↓ Damaging the surface and tools

MUST AVOID BUE!!

Poor quality of SURFACE! → Bad geometry! not well controlled!

→ this is connected to low cutting speed V_c

↓
you have to reach V_c to avoid adhesion of material, enough high V_c !



MECHANISM OF CHIP FORMATION

Chip breaker

⊗ you shape
RAKE so that,
after deformation,
IT breaks the
CHIP

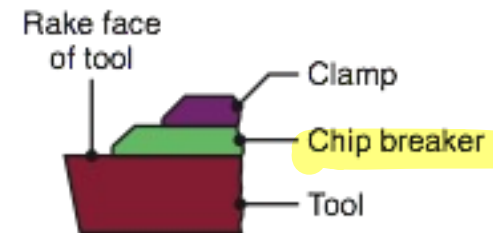
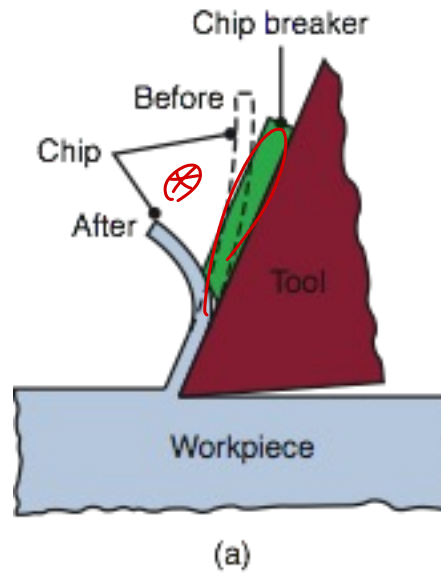
↓
additional deformation
WITH CHIP BREAKER

↓
break the continuous
chip in small pieces

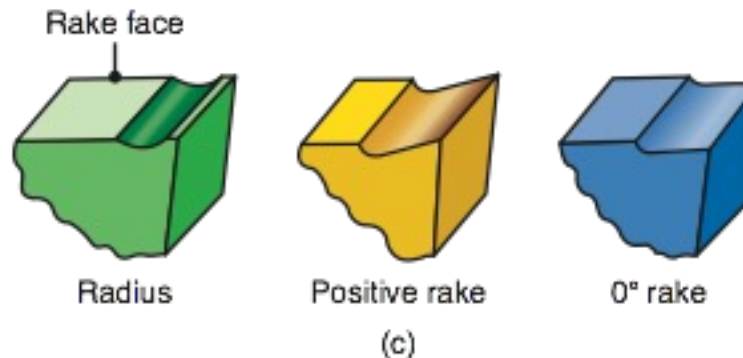
Grooves as
chip breaker

To EASY CHIP Removal

↓ try to deform chip, to
bring chips to fracture



shape rake face to have
further deformation on
chip to break it



MODEL OF CHIP FORMATION

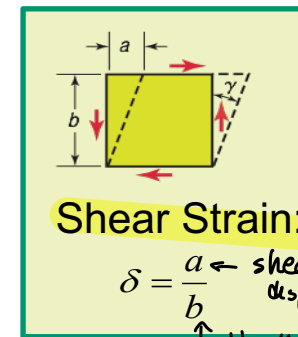
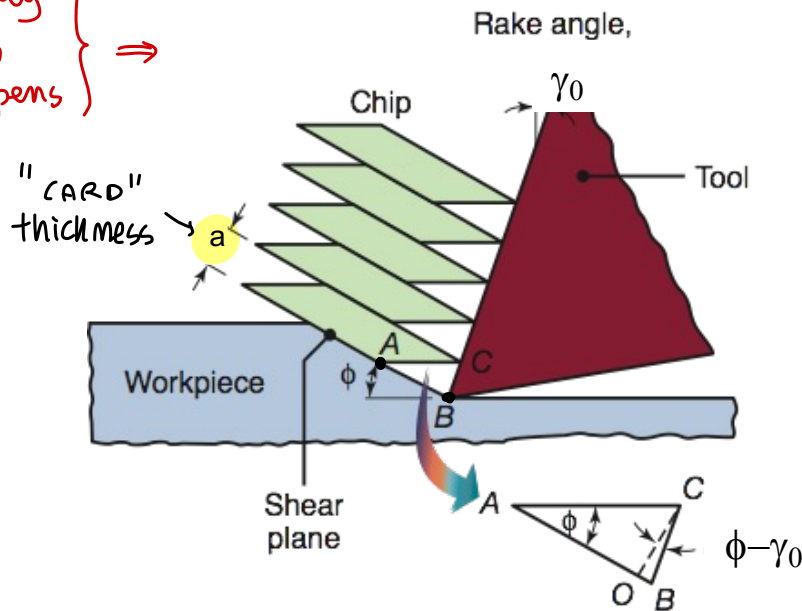
simplest MODEL to describe CHIP FORMATION

old good model

Card deck model of chip formation (Pijspanen, 1937)

Simple way
to look on
what happens $\Rightarrow$

for $a \rightarrow \alpha$
you have
the real
condition



Shear Strain:

$$\delta = \frac{a}{b} \leftarrow \begin{array}{l} \text{shear direction} \\ \text{displacement} \\ \uparrow \text{thickness} \Rightarrow \end{array}$$

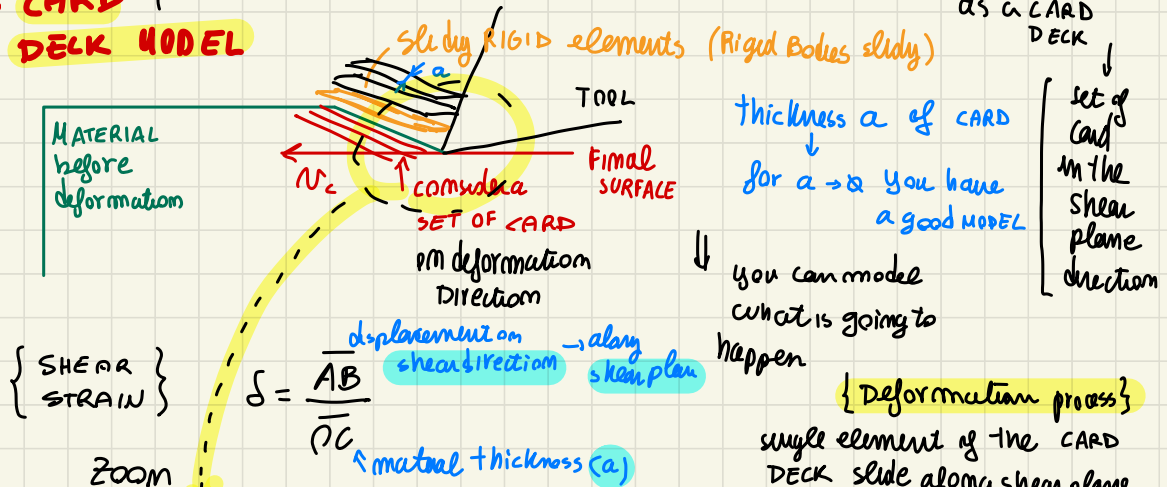
from that
model you can
describe what
happens --

Cutting ratio $r_c = \frac{h_D}{h_{ch}} = \frac{\sin \phi}{\cos(\phi - \gamma_0)}$

Shear strain $\delta = \frac{\overline{AB}}{\overline{OC}} = \frac{\overline{AO}}{\overline{OC}} + \frac{\overline{OB}}{\overline{OC}} = \cot \phi + \tan(\phi - \gamma_0)$

• **CARD** → representation of chip formation considering material before removal as a CARD DECK

DECK MODEL



{ Deformation process }

single element of the CARD DECK slide along shear plane and along RAKE face (SLIDING)

along RAKE face

$$\delta = \frac{\overline{AB}}{\overline{OC}} = \frac{AO}{OC} + \frac{OB}{OC} =$$

$$= \cot \phi + \tan(\phi - \gamma_0)$$

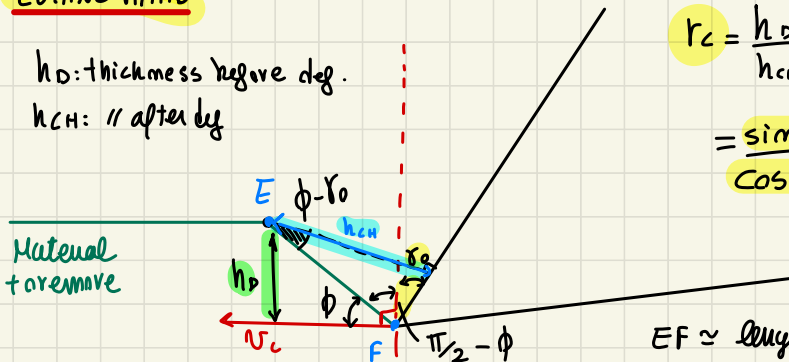
To evaluate ϕ evaluate, we can find shear strain δ

shear STRAIN related to main system angles

Chosen by us

CUTTING RATIO

h_0 : thickness before def.
 h_{CH} : " after def.

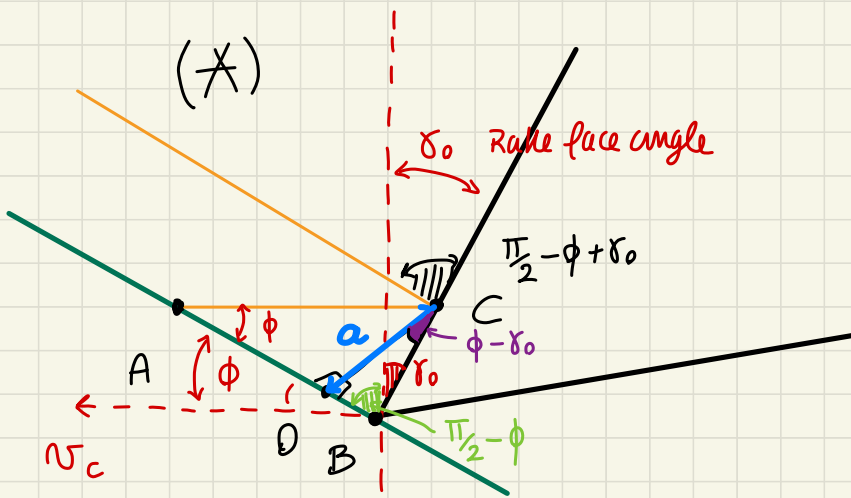


$$r_c = \frac{h_0}{h_{CH}} = \frac{EF \sin \phi}{EF \cos(\phi - \gamma_0)} =$$

$$= \frac{\sin \phi}{\cos(\phi - \gamma_0)}$$

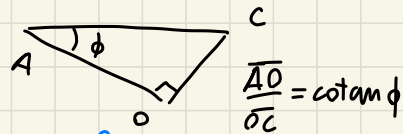
related only on the angles of the system!

$EF \approx$ length of shear plane
 IN CROSS-SECTION → $\perp$ to cutting edge



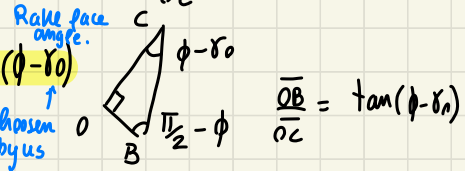
shear strain

SLIDING: AB
THICKNESS: OC



$$\delta = \frac{AB}{OC} = \frac{AO}{OC} + \frac{OB}{OC} = \cotan(\phi) + \tan(\phi - \delta_0)$$

↑ shear angle ↑ chosen by us



Card deck model

shear strain related
to the main angle of the system

the shear

angle ϕ → it is unknown a priori → angle of shear plane..
(unknown)

$$\rightarrow \sin \phi = r_c \cos(\phi - \gamma_0) = r_c [\cos \phi \cos \gamma_0 + \sin \phi \sin \gamma_0] =$$

FROM
CUTTING
RATIO
formula

$$\hookrightarrow r_c \cos \phi \cos \gamma_0 = \sin \phi - r_c \sin \phi \sin \gamma_0$$

↑ trigonometrical
formula

$$(1 - r_c \sin \gamma_0) \frac{\sin \phi}{\cos \phi} = r_c \cos \phi \cos \gamma_0$$

dividing by $\cos \phi$

way to
evaluate
the ϕ Angle!

$$\tan \phi = \frac{r_c \cos \gamma_0}{1 - r_c \sin \gamma_0}$$

FIXED

• IF we can evaluate
 r_c experimentally
and γ_0 known (fixed by us)

you can evaluate ϕ
experimentally (estimate ϕ)

CONSIDERING:

$$\bullet \bar{V}_{CHIP} = \text{constant}$$

$$b_D = b_{CH}$$

NO WIDTH

CHANGE $\rightarrow$ NO STRESS/STRAIN on that situation
(PLANAR ORTHOGONAL CUTTING)

$$h_D \cancel{b_D} \ell_D = h_{CH} \cancel{b_{CH}} \ell_{CH}$$

$$h_D \ell_D = h_{CH} \ell_{CH}$$

$$\rightarrow r_c = \frac{h_D}{h_{CH}} = \frac{\ell_{CH}}{\ell_D}$$

(for this, because $\ell_{CH} < \ell_D$
 $\rightarrow h_{CH} > h_D$)

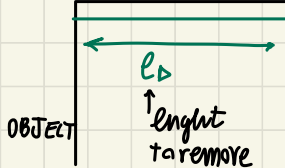
experimentally

chip removed
measured
able

$\hookrightarrow$ to estimate

r_c
 $\downarrow$
so ϕ estimation

easily to
measure on
a system



chip before
deformation $\rightarrow$ (to remove)
all will be removed

make experiment to
find ϕ

MODELS to find ϕ , use experiment

Data to validate this models!

(use experimental Data to understand what happen)

using the equation $h_D \ell_D = h_{CH} \ell_{CH}$ discuss

ISSUE for kinematic variables

$\rightarrow$ we have tool moving along v_c (relative motion) + other motions

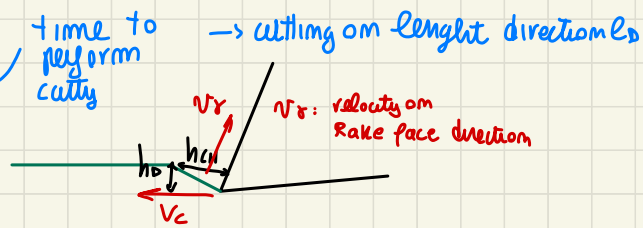
- chip movement (flow along rake)
- relative motion along shear plane



characterize
kinematic variables...

from $\frac{h_0 v_c}{t_c} = \frac{h_{CH} v_r}{t_c}$

dividing by $\left\{ \begin{array}{l} \text{cutty} \\ \text{time} \end{array} \right\}$



for MASS CONTINUITY

$$h_0 v_c = h_{CH} v_r$$

cause $\frac{h_0}{h_{CH}} < 1 \Rightarrow v_c > v_r$

BUT

$$v_c = \frac{h_0}{h_{CH}} = \frac{v_r}{v_c} = \frac{\sin \phi}{\cos(\phi - \delta_0)}$$

as found before...

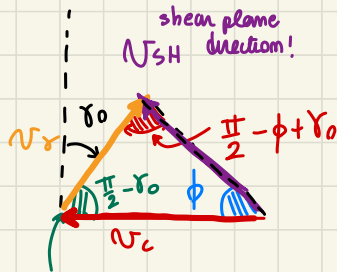
$$\frac{v_r}{\sin \phi} = \frac{v_c}{\cos(\phi - \delta_0)} = \frac{v_c}{\sin(\frac{\pi}{2} - \phi + \delta_0)} = \frac{v_{SH}}{\cos(\delta_0)}$$

can be evaluate by ϕ

by evaluating ϕ we have v_r, v_c

trigonometric equivalence

known $\sin(\frac{\pi}{2} - \delta_0)$



v_{SH} : Velocity along shear plane

full KINEMATIC DESCRIPTION

$$\pi - (\frac{\pi}{2} - \phi + \delta_0) - \phi = \frac{\pi}{2} - \delta_0$$

rule of triangle $\rightarrow$ rule of sign

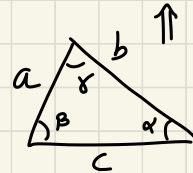
one side denoted $\sin(\text{opposite angle}) = \text{constant}$

known, choose by us

$$\frac{v_r}{\sin(\phi)} = \frac{v_c}{\cos(\phi - \delta_0)} = \frac{v_{SH}}{\cos(\delta_0)}$$

IF we are able to

evaluate $\phi \Rightarrow$ we can have clue about velocity along shear plane (v_{SH}) and along rake (v_r)



$$\frac{a}{\sin(\alpha)} = \frac{b}{\sin(\beta)} = \frac{c}{\sin(\gamma)}$$

$\hookrightarrow$ Full kinematic description!



Cutting ratio

$$r_c = \frac{h_D}{h_{ch}}$$

$$r_c = \frac{h_D}{h_{ch}} = \frac{b_{ch} \cdot l_{ch}}{b_D \cdot l_D} = \frac{l_{ch}}{l_D} \quad l_{ch} = \text{chip length}$$

Constant chip volume:

$$h_D \cdot b_D \cdot l_D = h_{ch} \cdot b_{ch} \cdot l_{ch}$$

Orthogonal cutting:

$$(b_D = b_{ch} = b)$$

$$r_c = \frac{h_D}{h_{ch}} = \frac{\overline{AB} \sin \phi}{\overline{AB} \cos(\phi - \gamma_0)} = \frac{\sin \phi}{\cos(\phi - \gamma_0)} \Rightarrow \tan \phi = \frac{r_c \cos \gamma_0}{1 - r_c \sin \gamma_0}$$



Velocity Diagram

Mass continuity: $v_c \cdot h_D = v_\gamma \cdot h_{ch}$

Since $h_{ch} > h_D \Rightarrow v_\gamma < v_c$.

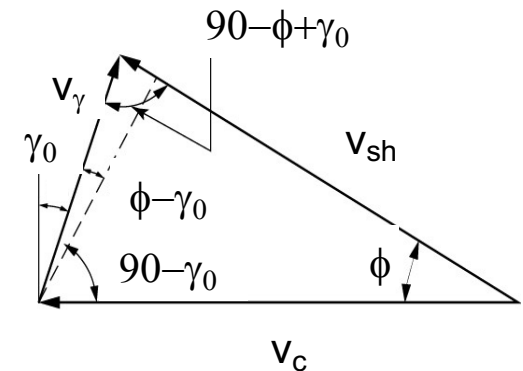
It follows that:

$$v_\gamma = v_c \frac{h_D}{h_{ch}} = v_c r_c = v_c \frac{\sin \phi}{\cos(\phi - \gamma_0)}$$

From the velocity diagram:

$$\frac{v_c}{\cos(\phi - \gamma_0)} = \frac{v_{sh}}{\cos \gamma_0} = \frac{v_\gamma}{\sin \phi}$$

Where v_{sh} is the speed in the shear plane.



Dynamic issue on chip formation

Forces can be described on the plane

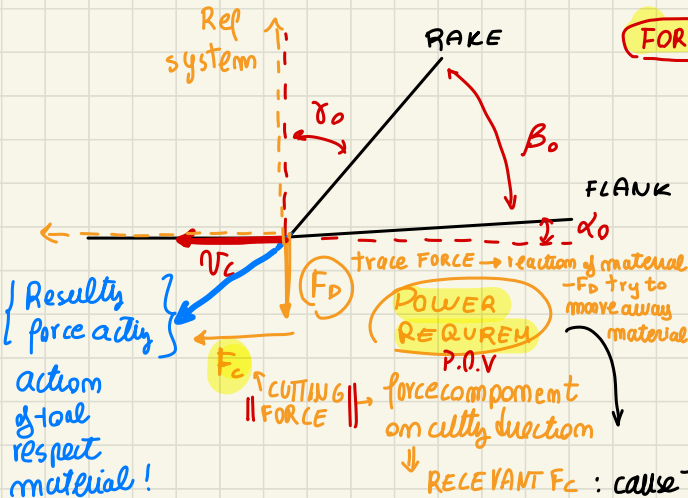
enables us to have chip formation
 ↓ More possible decomposition...

classical decomposition of

$F \rightarrow$ one component along V_c
 one orthogonal to V_c

keep tool in correct position

contrast force to material resistance in compression
 keep good position, F_D



ON ORTHOGONAL cutting power related only to CUTTING force

$$P = \vec{F} \cdot \vec{V} = F_c \cdot V_c + F_D \cdot V_D$$

$$\approx F_c \cdot V_c = P_c$$

(to evaluate Requirement x machining)

Velocity in direction ORTHOGONAL keep tool in position! $V_D = 0$ no move away

When PLANNING parameters should be chosen so that (influencing F_c ...)

$$P_c \leq \eta P_d$$

plug on the current...

depends on electrical power possible

$$0.75 \leq \eta \leq 0.95$$

less power usable

Typical Range of efficiency

(Net Power Usable)

MAX possible power due to efficiency limit that we can use for machining

• another possible decomposition of F

useful for friction

$$\mu = F_r / F_{rN}$$

β : FRICTION ANGLE $\rightarrow \mu = \tan \beta$

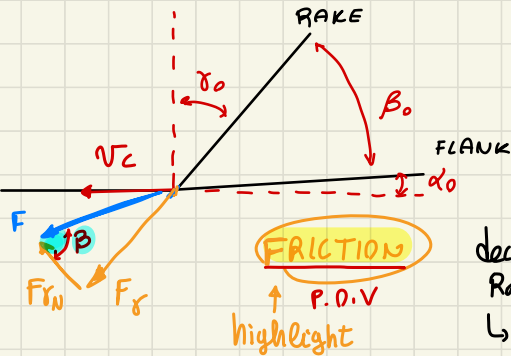
(Decompose the force)

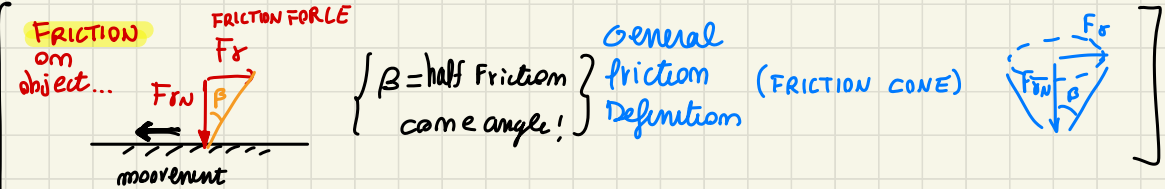
decompose along

Rake face and normal to that

to have information about FRICTION

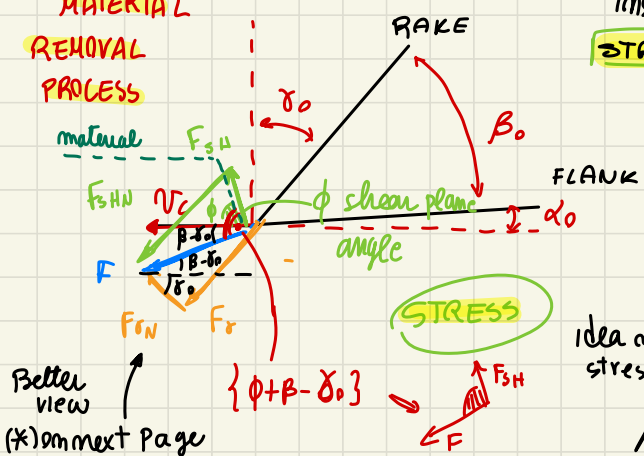
secondary shear zone due to FRICTION to find friction coefficient...





$\beta = \text{half Friction cone angle!}$

MATERIAL REMOVAL PROCESS



/// this decomposition instead is to find the STRESSES! $\{F_{SH}, F_{SHN}\}$ decompose along shear plane for TRUE STRESS situation

$$\tau_{SH} = \frac{F_{SH}}{A_{SH}} \text{ shear stress}$$

$$\sigma_{SH} = \frac{F_{SHN}}{A_{SH}} \text{ Normal stress}$$

of chip formation stress

idea of TRUE stress along shear plane $\rightarrow$ responsible of chip form.

All decompositions $\rightarrow$ STRESS

FRICTION

$$\begin{cases} F_r = F \sin \beta \\ F_{rN} = F \cos \beta \end{cases}$$

POWER REQ.

$$\begin{cases} F_c = F \cos(\beta - \alpha_0) \\ F_D = F \sin(\beta - \alpha_0) \end{cases}$$

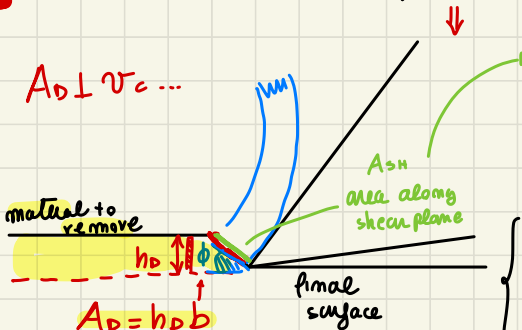
$$\begin{cases} F_{SH} = F \cos(\phi + \beta - \alpha_0) \\ F_{SHN} = F \sin(\phi + \beta - \alpha_0) \end{cases}$$

F can be measured, α_0, β fixed (FRICTION) $\rightarrow$ only ϕ MISSING! $\rightarrow$ we have to find it to understand what happens

chosen by us (given material condition $\rightarrow$ known $\mu \Rightarrow$ so β known)

$\Downarrow$ we should be able to PREDICT what happens $\rightarrow$ MODELS!

SO... CONSIDERING a particular situation



$$A_{SH} \sin \phi = A_D$$

$$A_{SH} = A_D / \sin \phi$$

depends on shear plane angle

from here: STRESS:

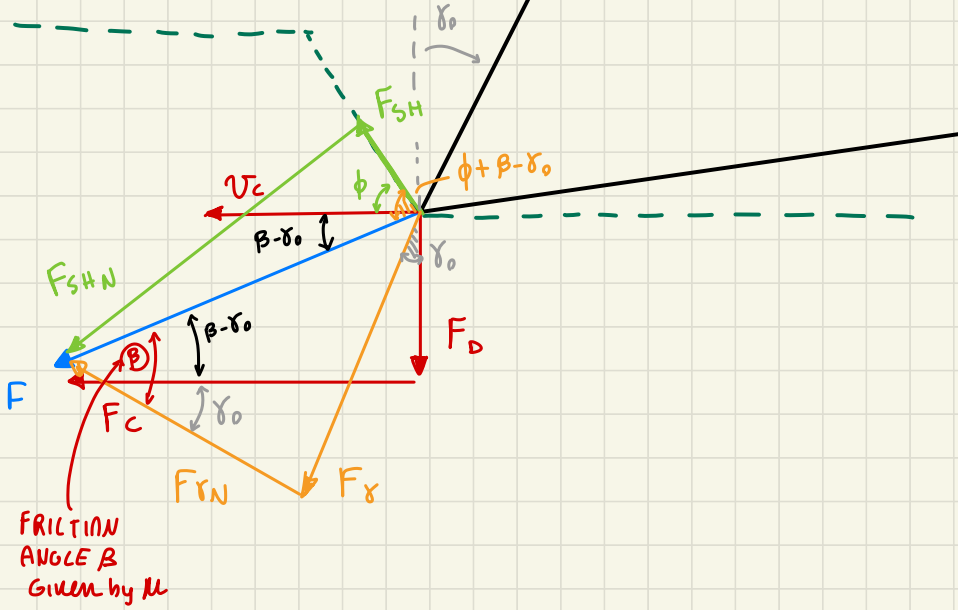
$$\tau_{SH} = F_{SH} / A_{SH} = F \cos(\phi + \beta - \alpha_0) / (A_D / \sin \phi) \text{ SHEAR}$$

$$\sigma_{SH} = F_{SHN} / A_{SH} = F \sin(\phi + \beta - \alpha_0) / (A_D / \sin \phi) \text{ NORMAL}$$

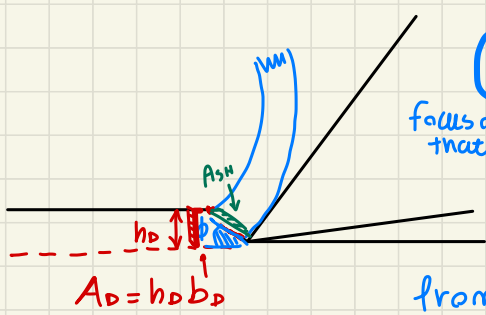
Cross section Area of what we are going to remove!

(*)

material



shear force acts on shear plane



focus on that

$$\tau_{SH} = (F_{SH}) / A_{SH} = F \cos(\phi + \beta - \delta_0) / (A_0 / \sin \phi) (*)$$

IMPORTANT relevant for CHIP FORMATION

also to predict

F_C (CUTTING FORCE) $\rightarrow$ so the POWER! for POWER REQ

we want to link shear stress τ_{SH} to cutting force F_C

from: $F_C = F \cos(\beta - \delta_0)$

$$\hookrightarrow F = F_C / \cos(\beta - \delta_0)$$

$$\tau_{SH} = \frac{F_C}{\cos(\beta - \delta_0)} \frac{\cos(\phi + \beta - \delta_0) \sin \phi}{A_D} \quad \text{CUTTING GEOMETRY}$$

so we can predict also P_C

From this equation
We evaluate F_C :

$$F_C = \frac{A_D \tau_{SH} \cos(\beta - \delta_0)}{\sin(\phi) \cos(\phi + \beta - \delta_0)}$$

link what happen along shear plane with other information! about cutting

To estimate F_C

- We know A_D
- we can evaluate τ_{SH}
- should know friction angle β (geometry + material + ecc.) + lubricant condition
- geometry δ_0
- $\rightarrow$ ϕ to find ϕ ?! $\Rightarrow \tau_{SH}$ can be determined (related to material)

HOW TO PREDICT that angle?! (*)

to try to material "destruction" τ_{SH} can be estimated

(*) TO EVALUATE F_C ...

ESTIMATION? $\rightarrow$ Looking on materials, est. of ways to find model estimation of ϕ

We need to find a way to estimate ϕ

- From first simplest model $\rightarrow$ develop a theory of estimation of ϕ

LOTS of articles scientific models of estimation

PIJS PAKEN by considering as Hypothesis: NO STRAIN HARDENING

τ_{max} without strain we keep removing material, without increasing τ ... no strain hard, so no required τ deformation continue!

simple model one of the FIRST

(constant properties of material)

- NO BUE (Buildup Edge)
- NO Friction

τ_{SH} as maximum required for fracture $\tau_{SH} = \tau_{MAX}$ NO MORE τ INCREMENT, IS MAXIMUM

Max τ along shear plane

(all deformation on shear plane, when apply max stress on shear plane)

from hypothesis $\Rightarrow$ $\phi : \min \delta$ } ϕ so as minimum shear strain δ } shear plane position to minimize

considered [FROM OUR objective:]

$\delta = \cot \phi + \tan(\phi - \gamma_0) \leftarrow$ shear strain

$$\frac{\partial \delta}{\partial \phi} = 0$$

$$\Leftrightarrow -\frac{1}{\sin^2 \phi} + \frac{1}{\cos^2(\phi - \gamma_0)} = 0$$

limited model
(NO FRICTION)

$$\frac{\sin^2 \phi - \cos^2(\phi - \gamma_0)}{\sin^2 \phi \cos^2(\phi - \gamma_0)} = 0$$

$$\sin^2 \phi = \cos^2(\phi - \gamma_0) \xrightarrow{\text{trigonometric relation}} \sin^2 \phi = \cos^2\left(\frac{\pi}{2} - \phi + \gamma_0\right)$$

$$\phi = \frac{\pi}{2} - \phi + \gamma_0 \rightarrow 2\phi = \frac{\pi}{2} + \gamma_0 \rightarrow \phi = \frac{\pi}{4} + \frac{\gamma_0}{2}$$

so from that model, we get the required value of ϕ

simple model does NOT define well what happen

$$\phi = \frac{\pi}{4} + \frac{\gamma_0}{2}$$

related to tool geometry

approx valid, but do NOT properly explain what is really happening $\rightarrow$ neglect FRICTION

{PIRKHEN MODEL}

► considering a Different model, Relevant MODEL

ERNST & MERCHANT model

Hyp: $\phi : \min E_c$ } ϕ so that energy needed for cutting is the MIN

assumption from that hypothesis...

- NO STRAIN HARD
- NO BUE (Built up Edge)
- μ constant (consider constant friction)
- τ_{SH} constant (requiring constant stress) (similar to hyp. of max value of shear stress along shear plane)

MAIN ASSUMPTION

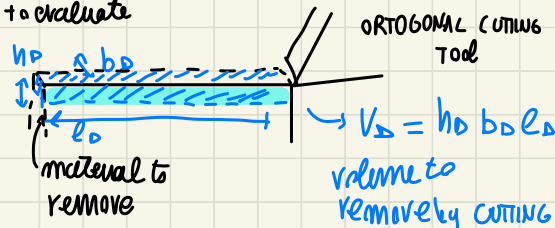
required energy for cutting: $E_c = F_c \cdot l_D$

cutting force along direction of motion

Minimize by

$$\frac{\partial E_c}{\partial \phi} = 0$$

(l_D or l_c as length of cutting)



$$E_c = F_c \cdot l_0 = \underbrace{l_0}_{\text{const. depend.}} \cdot \underbrace{A_D}_{\text{const. by tool}} \cdot \underbrace{\gamma_{SH}}_{\text{const. by tool}} \cdot \underbrace{\cos(\beta - \gamma_0)}_{\text{NOT function of } \phi}$$

by substituting the known quantity F_c

$$\downarrow \quad \underbrace{\sin \phi \cos(\phi + \beta - \gamma_0)}_{\text{constant numerator, so we can just analyze the denominator derivative}}$$

$$\frac{\partial E_c}{\partial \phi} = \frac{\partial}{\partial \phi} \left[\frac{1}{\sin \phi \cos(\phi + \beta - \gamma_0)} \right] = 0$$

$$= - \frac{\cos(\phi + \beta - \gamma_0) \cos \phi - \sin \phi \sin(\phi + \beta - \gamma_0)}{((\sin \phi) \cos(\phi + \beta - \gamma_0))^2} = 0$$

sum of cosine

$$\cos(\phi + \phi + \beta - \gamma_0) = \cos \pi/2 = 0 \quad \text{assumption by MIN energy requirements}$$

geometry of cutting

$$2\phi + \beta - \gamma_0 = \frac{\pi}{2} \Rightarrow \phi = \frac{\pi}{4} + \frac{\gamma_0}{2} - \frac{\beta}{2} \quad \left[\begin{array}{l} \text{E\&M model} \\ \text{method} \end{array} \right]$$

taking

E&M RESULT FROM HERE...

this model tell me that:
(discuss the Result, What about the FORCE?)

By substitution $(\beta - \gamma_0) = \frac{\pi}{2} - 2\phi$

$$F_c = \frac{A_D \gamma_{SH} \cos(\frac{\pi}{2} - 2\phi)}{\sin \phi \cos(\phi + \frac{\pi}{2} - 2\phi)} = \frac{A_D \gamma_{SH} \sin(2\phi)}{\sin \phi \sin \phi}$$

from the cutting FORCE expression

$$F_c = \frac{A_D \gamma_{SH} \cos(\beta - \gamma_0)}{\sin \phi \cos(\phi + \beta - \gamma_0)}$$

$$= A_D \frac{\gamma_{SH} 2 \cancel{\sin \phi} \cos \phi}{\cancel{\sin \phi} \cdot \sin \phi} = \frac{2 A_D \gamma_{SH}}{\tan \phi}$$

• FRICTION EFFECT:

$\mu \uparrow \rightarrow \beta \uparrow \rightarrow \phi \downarrow \rightarrow F_c \uparrow$ more cutting force! required

model that predict known real behaviour, so more friction lead to more cutting force required

• SHARPER TOOL

$\gamma_0 \uparrow \rightarrow \phi \uparrow \rightarrow F_c \downarrow$ less required force!
sharper tool by increase γ_0

(as expected!)
sharper tool
 $\downarrow$
easy cutting, less force required

When developing this Models ... considering $\mu \neq 0$ constant, FRICTION
 (E&M) well recognized prediction model } to predict FORCE... try to have
 analytical model, nice approx } accurate model

approx model: you need simulation, finite element analysis and discretize the material volume to perform simulation, describe material behaviour and tool interaction to simulate machining processes

for ex: Chip formation simulation .. Analytical model for good understanding + prediction

(FIRST understanding on what is happening)

↓ To predict what happens for real, we (relevant results obtained by an empirical model)
 use a MODEL empirical (from lot experiment) → enable to define PARAMETERS

that enables to find K_c : CUTTING PRESSURE [$\text{MPa} = \text{N}/\text{mm}^2$]

measured through $K_c \hat{=} F_c / A_D$ { by Definition }
 multiple experiment! / lubricant area before deform. → cut tool cross section

K_c depends on material / geometry cutting ecc... in CUTTING process
 (measure F_c on different condition... and from data we evaluate possible K_c values)

↓ knowing K_c ... ⇒ $F_c = K_c \cdot A_D$ MODEL VALID also in OBLIQUE CUTTING
 (find empirically)

and for prediction

improving ↓

$A_D = h_D b_D$ of the chip before deform.
 (particular value of cutting pressure)
 prevalent depends on work piece materials FROM EXPERIM. → specific cutting pressure

KRONENBERG relationship

that links K_c to particular values of the experiment

$K_c = \frac{K_{cs}}{h_D^x b_D^y}$ general model
 geometry influence
 x, y depends on tool material, comes from experiment

K_{cs} defined in a well defined situation with

$h_D = 1 \text{ mm}$
 $b_D = 1 \text{ mm}$ } $A_D = 1 \text{ mm}^2$
 "SPECIFIC" with h_D, b_D fixed

to predict FORCE

INDUSTRIAL empirical model

nowadays usually $y \approx 0$

(for tool material used $y=0$) $K_c \hat{=} \frac{K_{cs}}{(h_D)^x}$ WORKPIECE material
 TOOL material

NOW mainly K_c depends on WORKPIECE mat K_{cs} TOOL material x and GEOMETRY h_D

Good force prediction for machining operation thanks to planning this model

usually K_{cs}, x are material given by the tool maker { we need empirical values } K_{cs}, x

before deformation chip thickness (depends on GEOMETRY OF CUTTING)

- If making machine tools... I should know the forces! (INTERACTION)
(all forces components must be known)

- If using the machine → use Model of KRONENBERG
Just planning the operation is enough in industry to plan operation of machine

↓ good prediction...
simulation needed for understand interaction

Described Power Requirements!

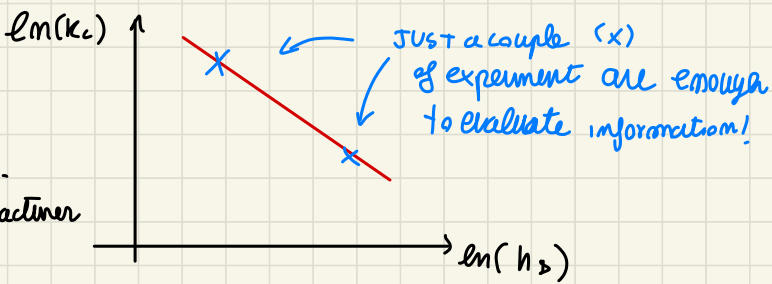
predicting F_c , we can evaluate feasibility of machine if $P_c \leq \eta P_c$
 NOTICE: on KRONENBERG Relation: $\uparrow$ found from P_c

CONSIDERATION usually we use on LOG form

$$K_c = \frac{K_{cs}}{h_D^x} \rightarrow \ln(K_c) = \ln(K_{cs}) - x \ln(h_D)$$

↓ from this "linear relation" it is a line

from linear log relationship..
the tool manufacturer gives you the parameters K_c, K_{cs}, x
↓



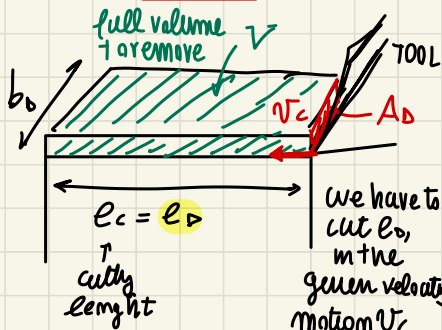
$\ln(K_c) = \ln(K_{cs}) - x \ln(h_D)$ it is a line on that plane

considering a possible application... (h_D) ⇒ Just a couple of experiment are enough to estimate the parameters (POWERFUL tool!)

Other issue: • Cutting Time

t_c

DISCUSS about TIME
CUTTING TIME = time in which tool is in contact with material

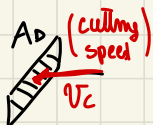


$$t_c = \frac{l_D}{V_c} = \frac{(b_0 b_D) l_D}{V_c} = \frac{V}{A_D V_c}$$

$$= \frac{V}{A_D V_c}$$

AD CHIP AREA TO REMOVE

(x) if considering all the operation time... first tool measurement, etc. → it is all MACHINING TIME t_m



$$A_D \cdot V_c = \left[\frac{\text{mm}^3}{\text{s}} \right] \text{ volume per unit of time}$$

(Defined) { MATERIAL REMOVAL RATE } MRR

VALID $V_{\text{machining}}$ process! (turning/milling/drilling)

general expression for any machining processes

$$t_c = \frac{V}{MRR}$$

this must be evaluated V kind of process

volume to remove, amount of material to remove

depends on CUTTING parameter $MRR = A_D \cdot V_c$ { geometry velocity

that we have to choose

• ENERGY for CUTTING (chip formation)

$$E_c = F_c \cdot l_D = \underbrace{K_c A_D l_D}_{F_c} = K_c b_D h_D l_D = K_c V$$

equally $E_c = P_c \cdot t_c = \dots = K_c V$ same result! We reach the same result!

$$K_c = \frac{E_c}{V}$$

$K_c \equiv$ cutting energy per unit of volume (specific energy) related to cutting

usually given in [MPa]

$$[K_c] = [\text{MPa}] = \frac{\text{N}}{\text{mm}^2} = \frac{(\text{N} \cdot \text{mm})}{\text{mm}^3} = [\text{J}] \leftarrow \frac{\text{Energy}}{\text{Volume}}$$

SPECIFIC CUTTING ENERGY

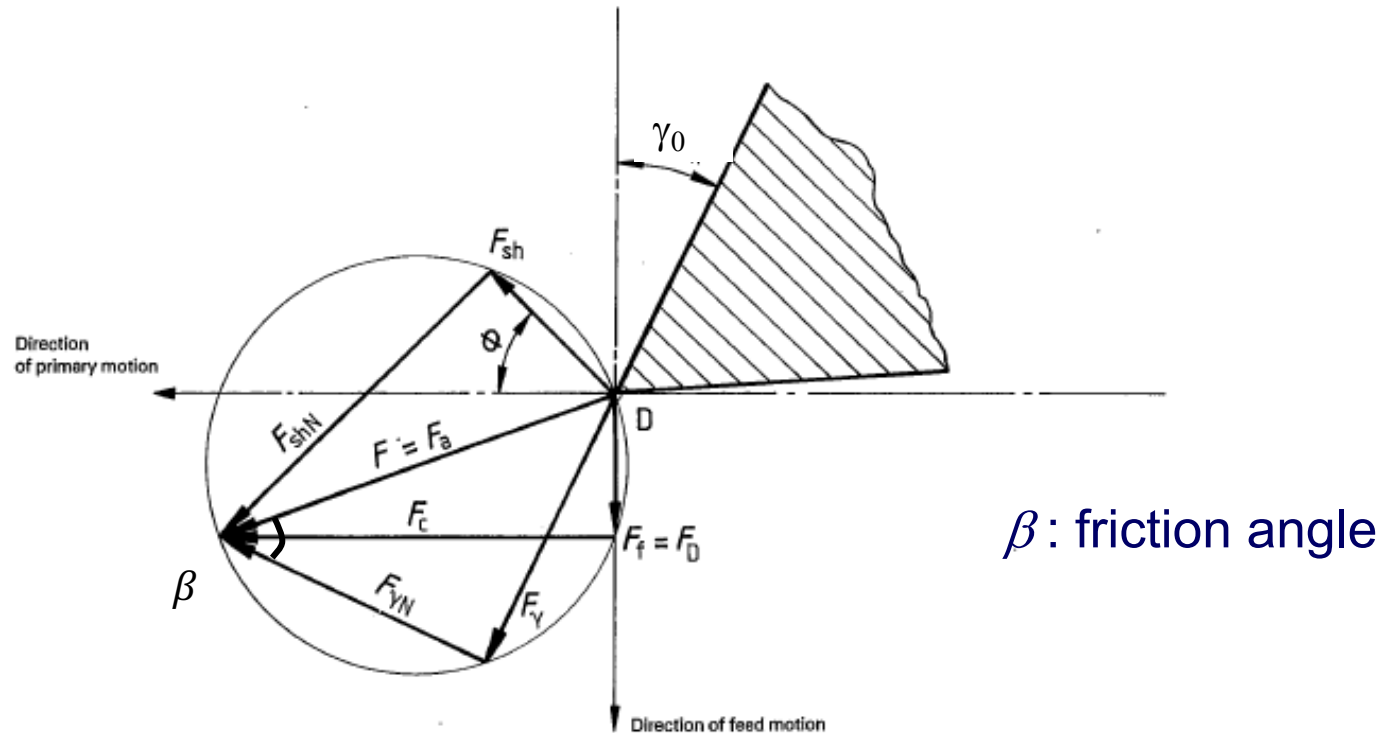
to properly predict $E_c \rightarrow$ energy requirement (the energetic issue nowadays is relevant)

K_c useful value!! K_c well known! easy to estimate $E_c = K_c V$

depending on material of workpiece (K_c) of tool (x) and geometry (h_D)

removing chip thickness

The force F represents the action of the tool on the material. It can be decomposed in different ways to fully characterize the tool-material interaction.



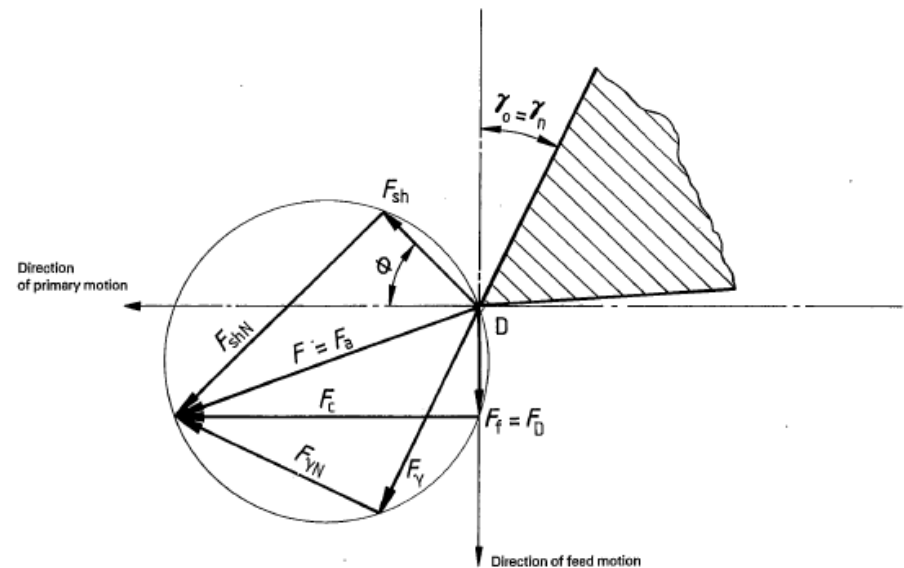
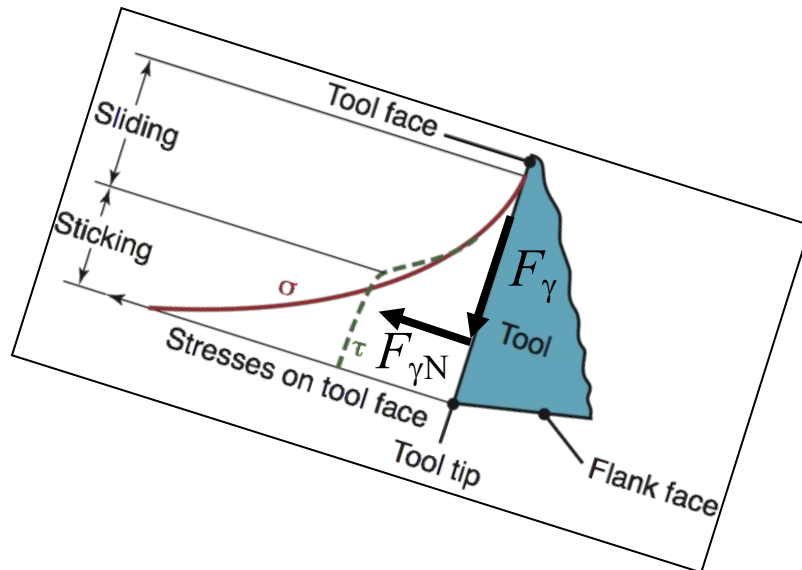


Force F_γ : parallel to the rake plane

Force $F_{\gamma N}$: normal to the rake plane

$$F_\gamma = F \sin(\beta)$$

$$F_{\gamma N} = F \cos(\beta)$$





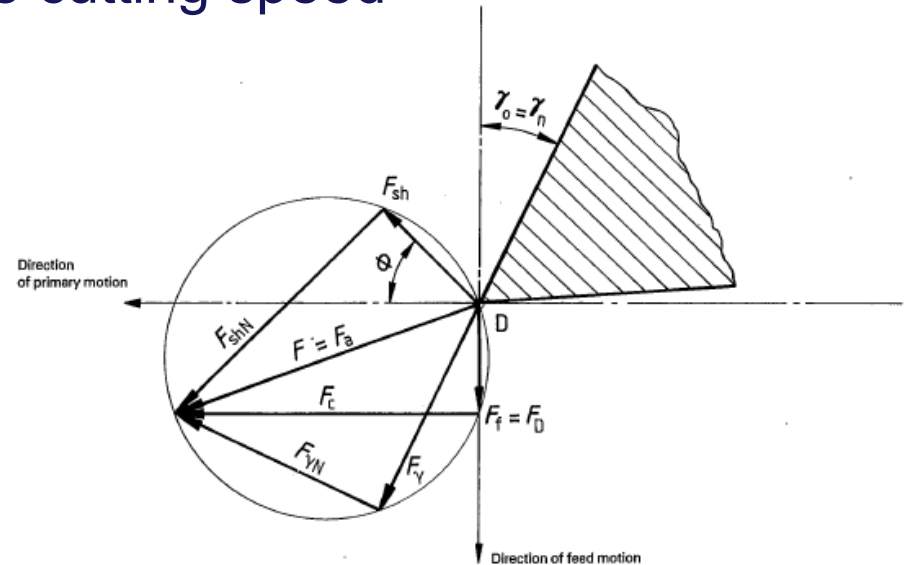
MODEL OF CHIP FORMATION

Cutting force (F_c), parallel to the cutting speed

Thrust force (F_D), normal to the cutting speed

$$F_c = F \cos(\beta - \gamma_0)$$

$$F_D = F \sin(\beta - \gamma_0)$$



These forces are not known, but they can be measured or predicted with mathematical models.



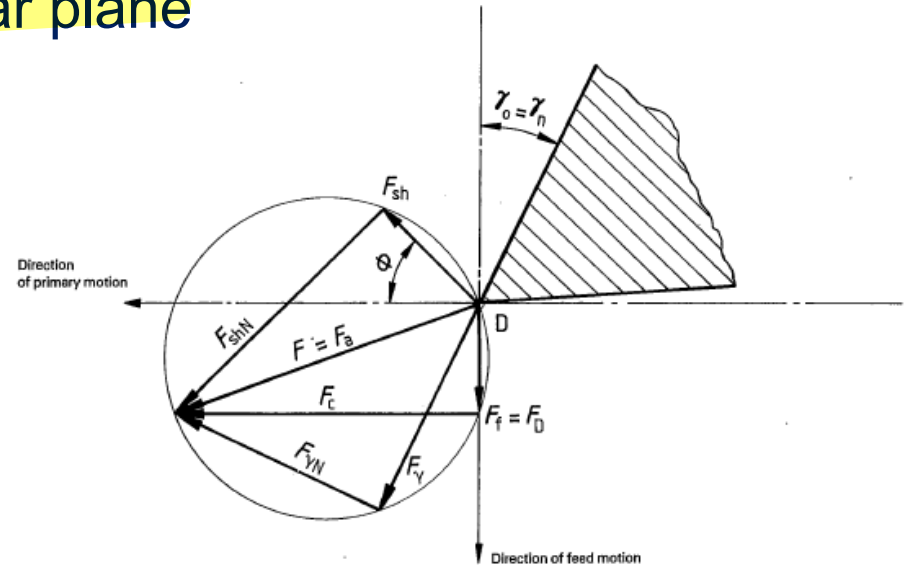
MODEL OF CHIP FORMATION

Force F_{sh} : parallel to the shear plane

Force F_{shN} : normal to the shear plane

$$F_{sh} = F \cos(\phi + \beta - \gamma_0)$$

$$F_{shN} = F \sin(\phi + \beta - \gamma_0)$$





MODEL OF CHIP FORMATION

$$\tau_{sh} = \frac{F_{sh}}{A_{sh}} = \frac{F_{sh} \sin \phi}{A_D} = \frac{F \cos(\phi + \beta - \gamma_0) \sin \phi}{A_D} = \frac{F_c \cos(\phi + \beta - \gamma_0) \sin \phi}{A_D \cos(\beta - \gamma_0)}$$

$$\sigma_{sh} = \frac{F_{shN}}{A_{sh}} = \frac{F_{shN} \sin \phi}{A_D} = \frac{F \sin(\phi + \beta - \gamma_0) \sin \phi}{A_D} = \frac{F_c \sin(\phi + \beta - \gamma_0) \sin \phi}{A_D \cos(\beta - \gamma_0)}$$

Parameters: $\gamma_0, \beta, \phi, \tau_{sh}$

- γ_0 is a property of the cutting tool
- β can be estimated:

$$\mu = \tan \beta = \frac{F_\gamma}{F_{\gamma N}} = \frac{F_D \cos \gamma_0 + F_c \sin \gamma_0}{F_c \cos \gamma_0 - F_D \sin \gamma_0} = \frac{F_D + F_c \tan \gamma_0}{F_c - F_D \tan \gamma_0}$$



$$F_c = F \cos(\beta - \gamma_0)$$

$$F_{sh} = F \cos(\phi + \beta - \gamma_0) \Rightarrow F = \frac{F_{sh}}{\cos(\phi + \beta - \gamma_0)}$$

$$F_{sh} = \tau_{sh} A_{sh} = \tau_{sh} \frac{A_D}{\sin \phi}$$



$$F_c = \frac{F_{sh} \cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)} = \tau_{sh} \frac{A_D}{\sin \phi} \frac{\cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)}$$



MODEL OF CHIP FORMATION

Pijspanen model

- No strain hardening
- No BUE (Built Up Edge)
- Plastic deformation begins when $\tau_{\max} = \tau_{\text{sh}}$
(elastic strain is not considered)
- No friction on the flank
- No friction between tool's rake and chip

Shear angle ϕ assumes a value that minimizes the shear strain.



Pijspanen model

$$\frac{\partial \delta}{\partial \phi} = \frac{\partial}{\partial \phi} [\cot \phi + \tan(\phi - \gamma_0)] = -\frac{1}{(\sin \phi)^2} + \frac{1}{(\cos(\phi - \gamma_0))^2}$$

$$\frac{(\sin \phi)^2 - (\cos(\phi - \gamma_0))^2}{[\sin \phi \cos(\phi - \gamma_0)]^2} = 0$$

$$(\sin \phi)^2 = (\cos(\phi - \gamma_0))^2 = \left[\sin \left(\frac{\pi}{2} - (\phi - \gamma_0) \right) \right]^2$$

$$\phi = \frac{\pi}{2} - (\phi - \gamma_0) \quad \Rightarrow \quad \phi = \frac{\pi}{4} + \frac{\gamma_0}{2}$$



MODEL OF CHIP FORMATION

Ernst & Merchant

- No strain hardening
- No BUE (Built Up Edge)
- No friction on the flank
- Constant τ_{sh}

Shear angle ϕ assumes a value that minimizes the cutting energy.



Ernst & Merchant

$$F_c = \tau_{sh} \frac{A_D}{\sin \phi} \frac{\cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)}$$

$$E_c = F_c L_c = \tau_{sh} \frac{A_D}{\sin \phi} \frac{\cos(\beta - \gamma_0)}{\cos(\phi + \beta - \gamma_0)} L_c$$

$$\frac{\partial E_c}{\partial \phi} = 0$$



Ernst & Merchant

$$\frac{\partial E_c}{\partial \phi} = \tau_{sh} A_D L_c \cos(\beta - \gamma_0) \frac{\partial}{\partial \phi} \left[\frac{1}{\sin \phi \cos(\phi + \beta - \gamma_0)} \right] = 0$$

$$\frac{\cos \phi \cos(\phi + \beta - \gamma_0) - \sin \phi \sin(\phi + \beta - \gamma_0)}{[\sin \phi \cos(\phi + \beta - \gamma_0)]^2} = 0$$

$$\cos(\phi + \phi + \beta - \gamma_0) = 0 = \cos \frac{\pi}{2}$$

$$2\phi + \beta - \gamma_0 = \frac{\pi}{2} \quad \Rightarrow \quad \phi = \frac{\pi}{4} + \frac{1}{2}(\gamma_0 - \beta)$$



Let's consider the Ernst & Merchant model result:

$$2\phi + \beta - \gamma_0 = \frac{\pi}{2} \rightarrow \beta - \gamma_0 = \frac{\pi}{2} - 2\phi$$

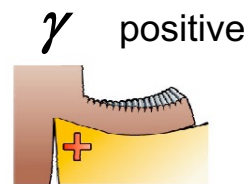
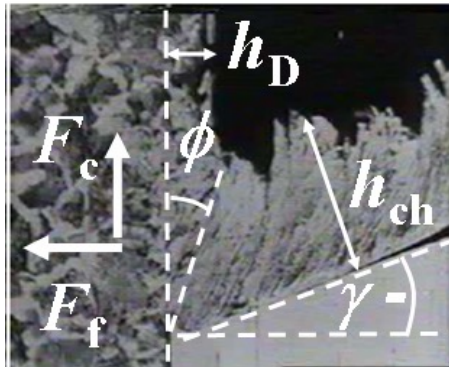
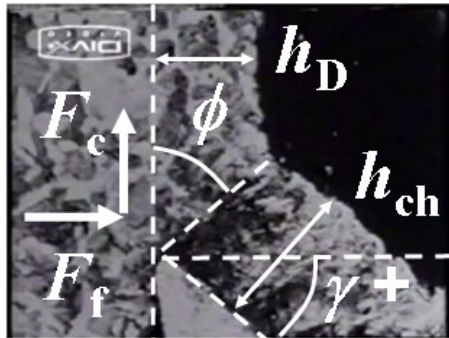
$$F_c = \frac{\tau_{sh} \cdot A_D \cdot \cos(\beta - \gamma_0)}{\sin \phi \cdot \cos(\phi + \beta - \gamma_0)} = \frac{\tau_{sh} \cdot A_D \cdot \cos(\frac{\pi}{2} - 2\phi)}{\sin \phi \cdot \cos(\phi + \frac{\pi}{2} - 2\phi)}$$

$$F_c = \frac{\tau_{sh} \cdot A_D \cdot \sin 2\phi}{\sin \phi \cdot \sin \phi} = \frac{\tau_{sh} \cdot A_D \cdot 2 \sin \phi \cos \phi}{\sin \phi \cdot \sin \phi}$$

$$F_c = \frac{2 \cdot \tau_{sh} \cdot A_D}{\tan \phi}$$



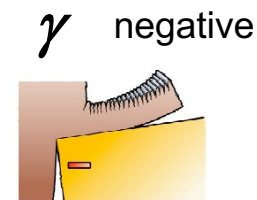
Influence of γ and β on forces



γ_0 ↑
 β ↓

$\beta - \gamma_0$ ↓
 ϕ ↑
 h_{ch} ↓

F_c ↓



γ_0 ↓
 β ↑

$\beta - \gamma_0$ ↑
 ϕ ↓
 h_{ch} ↑

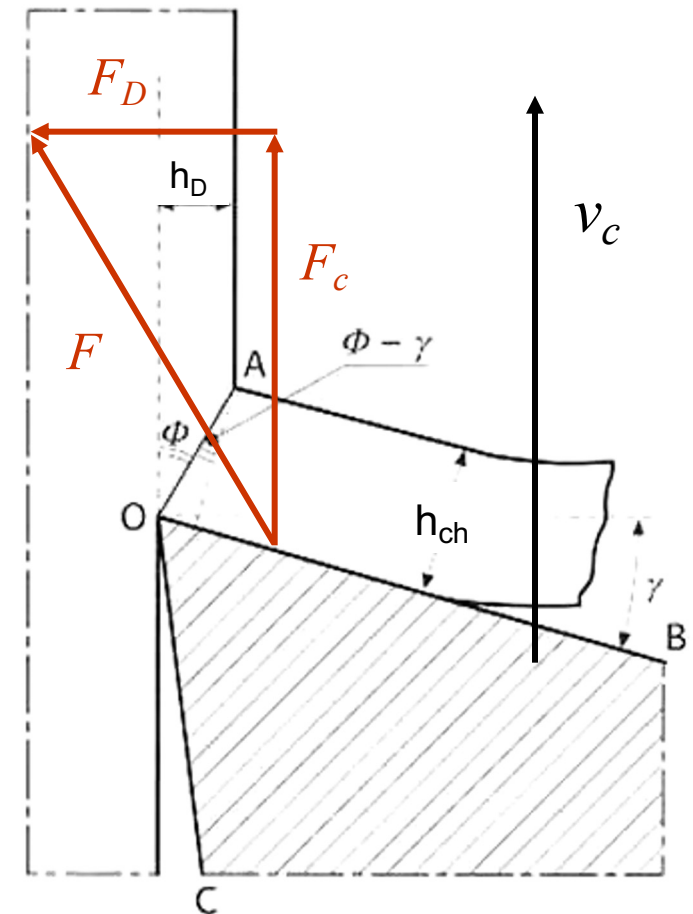
F_c ↑



F_c is the cutting force, whose direction is parallel to the cutting speed v_c .

However the total force F in orthogonal cutting may be considered as the composition of:

- Cutting Force: F_c
- Thrust Force: F_D



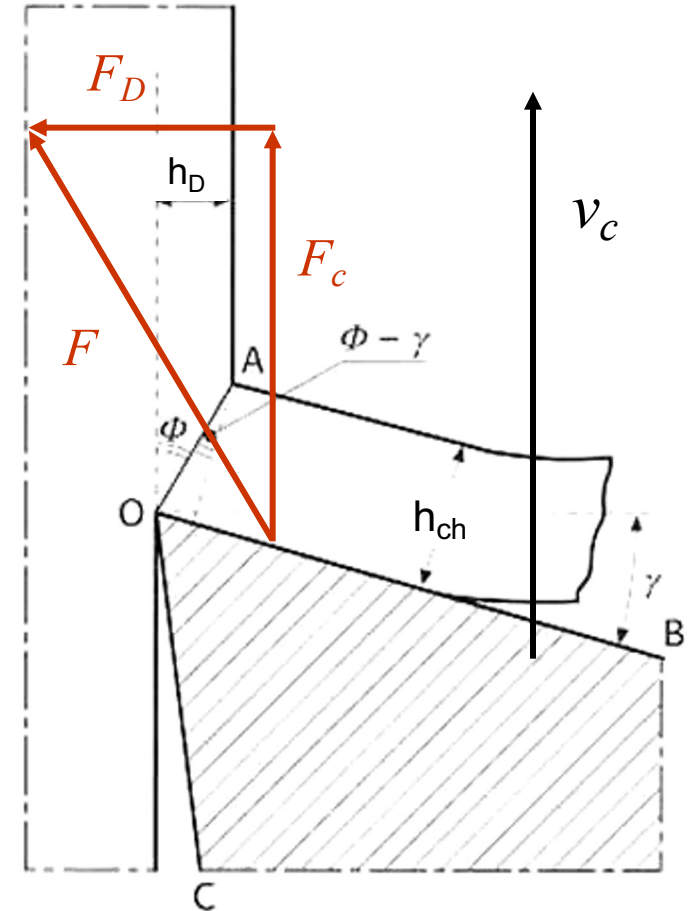


If $\vec{v}$ is the relative speed between tool and workpiece, the required power P is given by the scalar product:

$$P = \vec{F} \cdot \vec{v} = F_c v_c + F_D v_D$$

But, since $v_D = 0$, it follows that required power is the cutting power P_c :

$$P_c = F_c v_c$$





The cutting pressure k_c is defined from the relationship:

$$k_c \stackrel{\text{def}}{=} \frac{F_c}{A_D} \rightarrow F_c = k_c A_D$$

k_c depends on:

- A_D
- workpiece material
- tool material and geometry (γ_0 in particular)
- v_c
- lubrication conditions



Kronenberg relationship:

$$k_c = \frac{k_{cs}}{h_D^x \cdot b^y}$$

Where k_{cs} is the specific cutting pressure to remove a chip section of 1 mm² with $h_D = 1$ mm and $b = 1$ mm.

Typically, $y \cong 0$, then it follows that:

$$k_c = \frac{k_{cs}}{h_D^x}$$

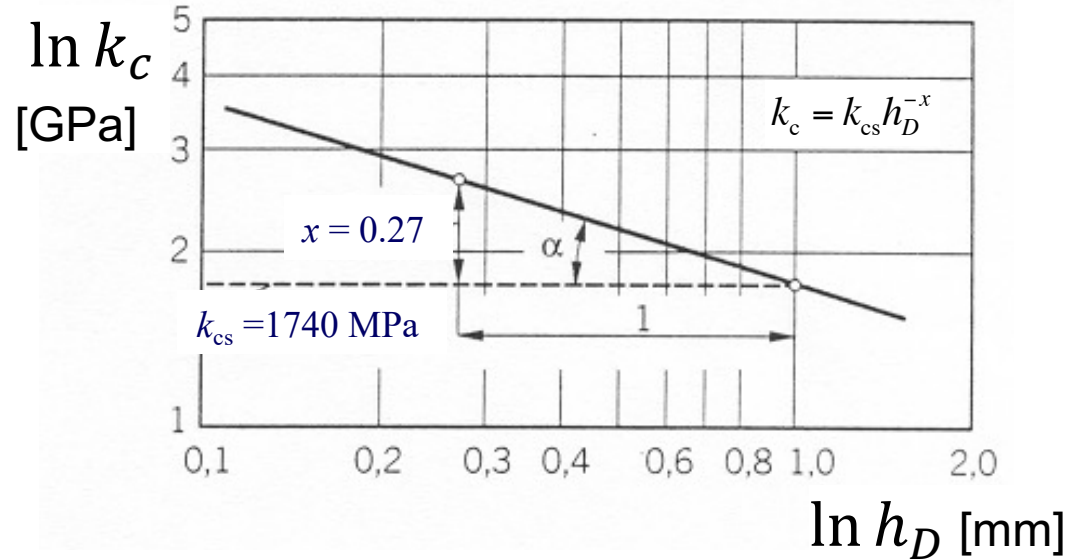
Note that:

- k_{cs} related to the material to remove;
- k_c decreases as h_D increases;
- x constant mainly related to the tool material.



$$k_c = \frac{k_{cs}}{h_D^x}$$

$$\ln k_c = -x \ln h_D + \ln k_{cs}$$



NB: k_{cs} is a value that you can find in books and catalogues



The power given by the motor is the absorbed electric power (P_e) reduced by the transformation efficiency (factor η).

Therefore, it follows that:

$$P_c \leq \eta \cdot P_e$$

In general $0,7 \leq \eta \leq 0,95$, while the maximum value P_e is a given characteristic of the machine.



CUTTING TIME AND MATERIAL REMOVAL RATE

In orthogonal cutting, the cutting time t_c can be computed as:

$$t_c = \frac{L_c}{v_c}$$

where L_c is the cutting length and v_c is the cutting speed.

It can also be computed as:

$$t_c = \left(\frac{h_D b}{h_D b} \right) \frac{L_c}{v_c} = \frac{V}{MRR}$$

where:

- V is removed material volume [mm³]
- $MRR = A_D \cdot v_c$ is the material removal rate [mm³ / min.]



In orthogonal cutting, the cutting energy E_c can be computed as:

$$E_c = F_c L_c = k_c A_D L_c = k_c \overbrace{h_D b L_c}^{\bar{V}} = k_c V$$

It can also be computed as:

$$E_c = P_c t_c = F_c v_c \overbrace{\left(\frac{V}{MRR} \right)}^{t_c} = k_c A_D v_c \frac{V}{A_D v_c} = k_c V$$

Therefore, the cutting pressure k_c can be also seen as the specific cutting energy $\left[\frac{N \cdot mm}{mm^3} \right]$ for the machining operation.